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Dery et al.

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(54) **ROOFTOP MOUNT FOR EXERCISE EQUIPMENT**

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Related U.S. Application Data

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(Continued)

(51) **Int. Cl.**
A63B 71/02 (2006.01)
A63B 71/00 (2006.01)

(52) **U.S. Cl.**
CPC **A63B 71/023** (2013.01); **A63B 71/0036** (2013.01)

(58) **Field of Classification Search**
CPC ... A63B 21/16; A63B 21/00047; A63B 17/00; A63B 71/0036; A63B 71/023;
(Continued)

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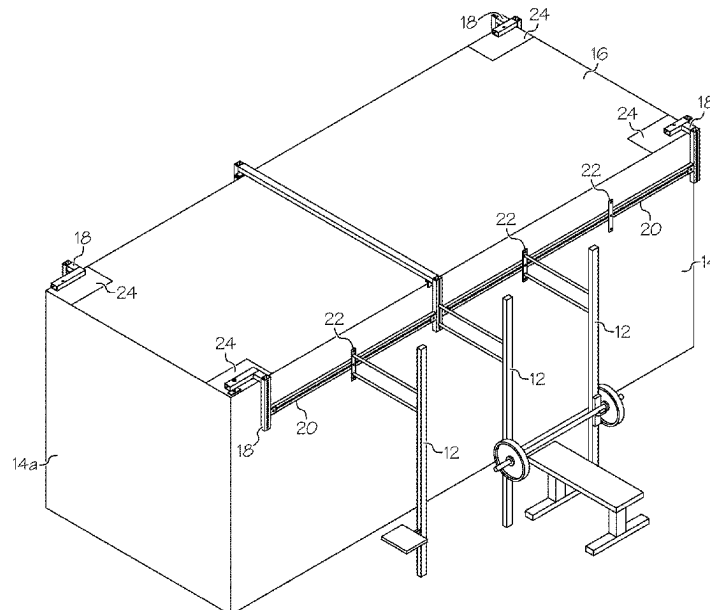
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Primary Examiner — Hilary L Gutman

(57) **ABSTRACT**

A rooftop mount device for connecting to a roof of a portable gym and for attaching exercise equipment is described. The portable gym is a portable container, such as a shipping container, having a top surface, or roof, to which the rooftop mount is attachable. The rooftop mount device includes connection members. The rooftop mount can also include one or more horizontal stabilization members and equipment attachment elements. The connection members are attached to a roof of the container. Exercise equipment is attached to the parts of the rooftop mount device. The rooftop mount provides proper height and stability for the bars, racks, stands, cages, rigs and riggings, and other exercise equipment that are attached to the container's roof so that the equipment does not move during use by an exerciser. Systems and kits are also described, which include a rooftop mount device, various exercise equipment, and a portable container.

16 Claims, 16 Drawing Sheets



Related U.S. Application Data

(60) Provisional application No. 62/795,308, filed on Jan. 22, 2019.

(58) **Field of Classification Search**

CPC . B60P 3/00; B60P 3/1008; B60R 9/08; B60R 11/00; B60R 2011/0042; B65D 88/121
USPC 296/24.3; 482/148
See application file for complete search history.

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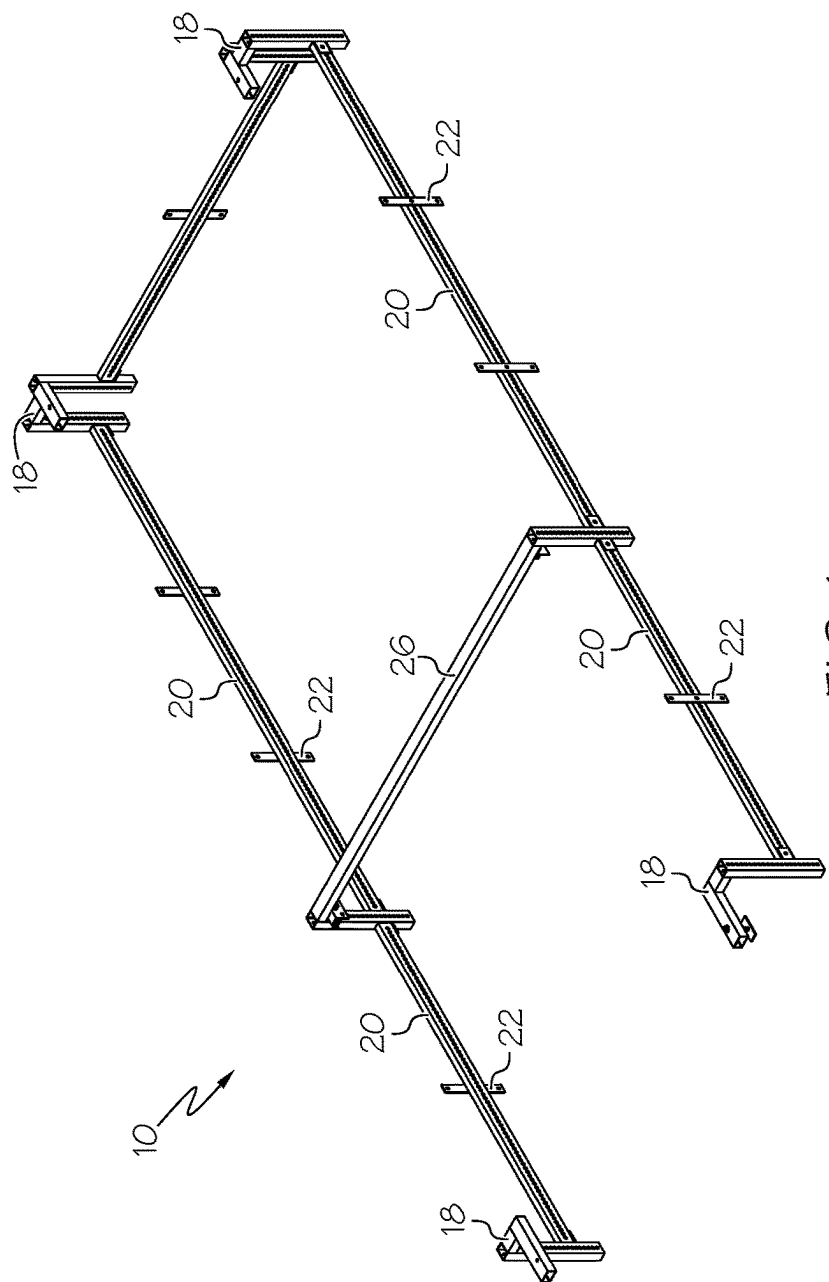


FIG. 1

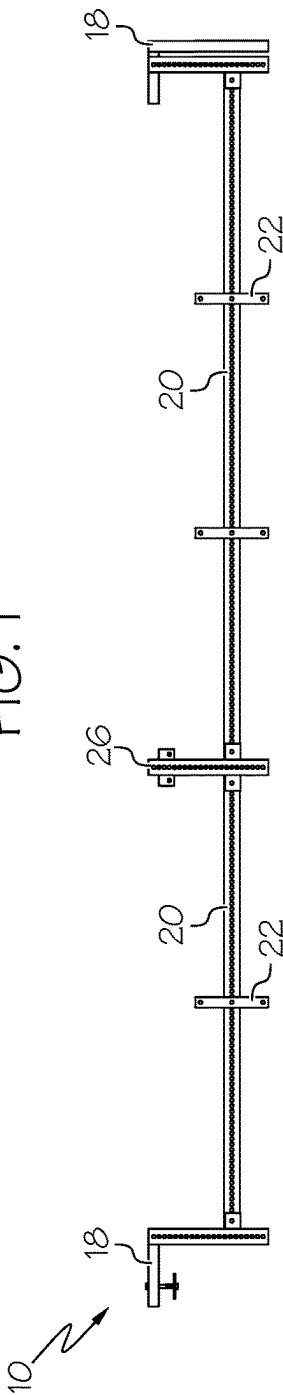


FIG. 2

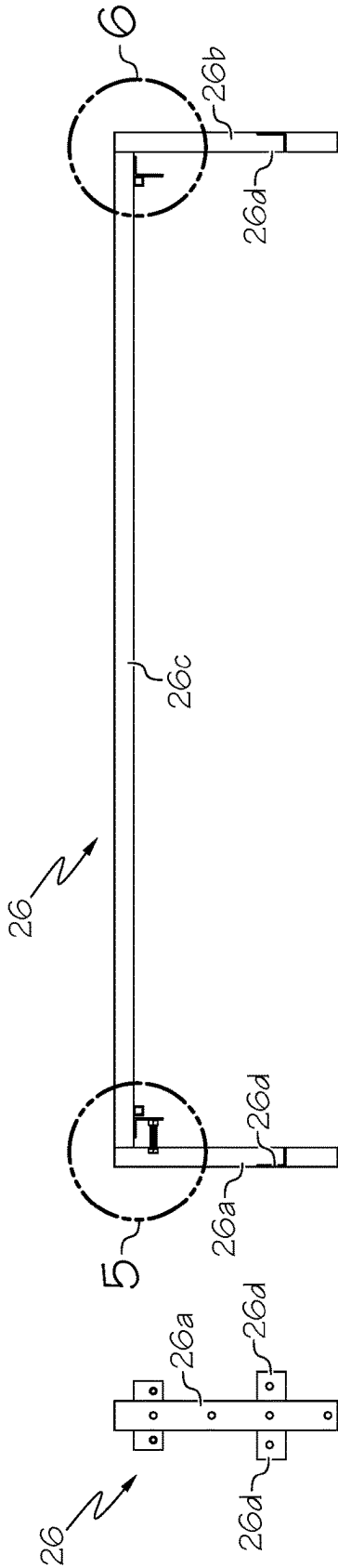


FIG. 4

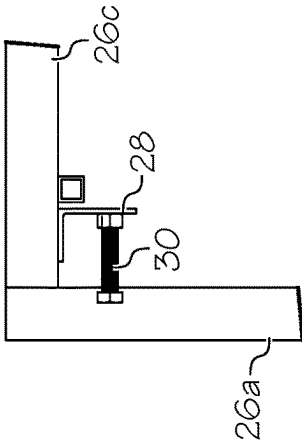
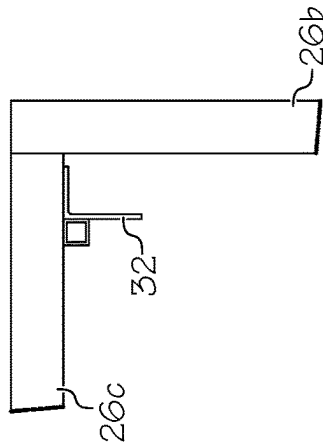


FIG. 6

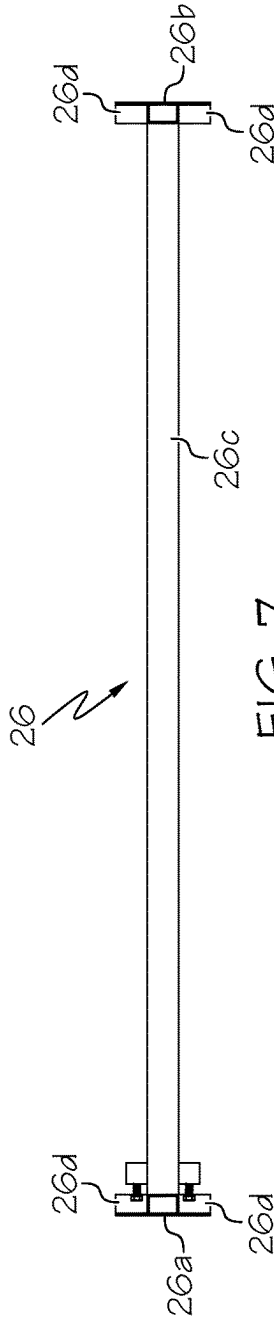


FIG. 7



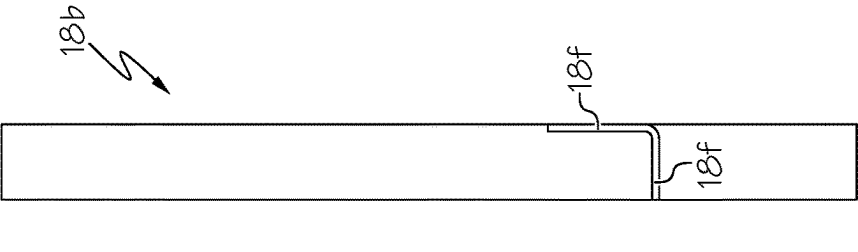


FIG. 8

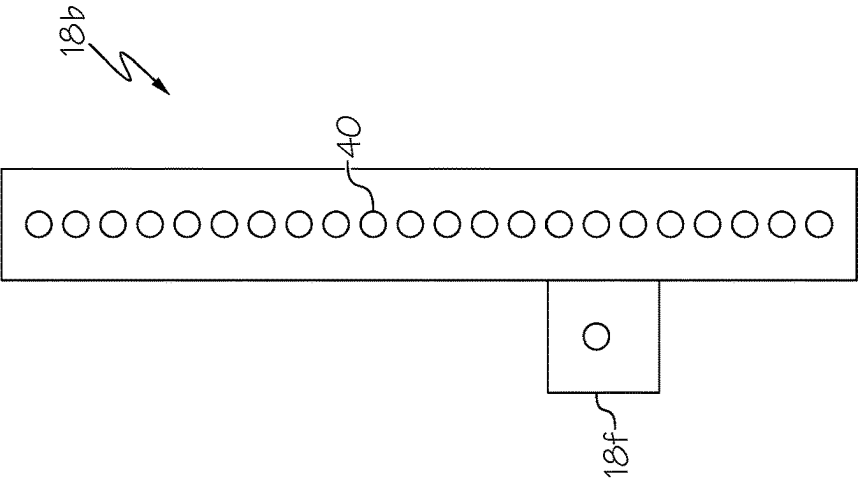


FIG. 9

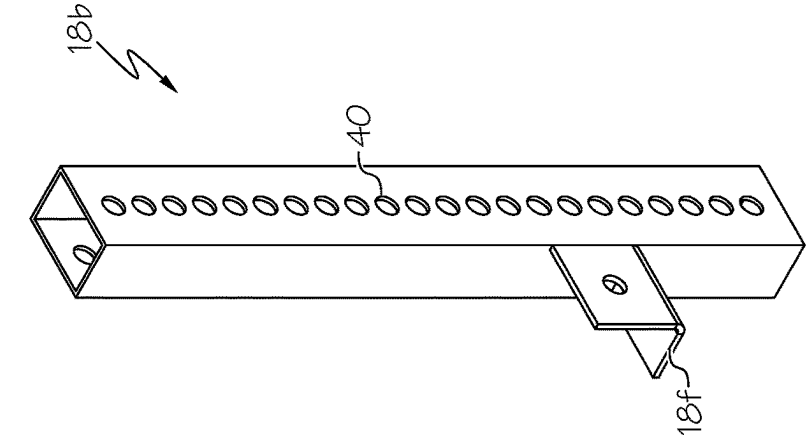


FIG. 10

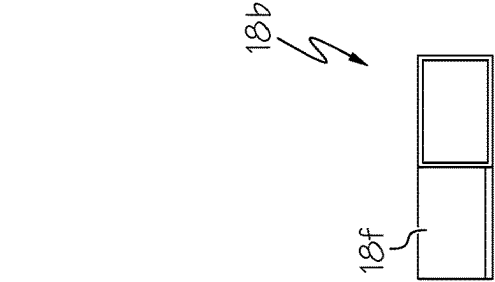


FIG. 11

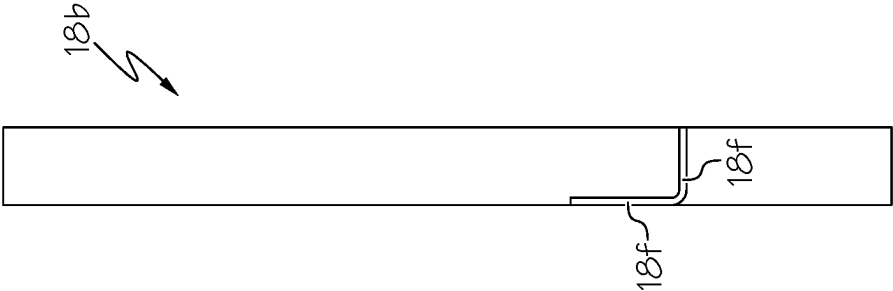


FIG. 14A

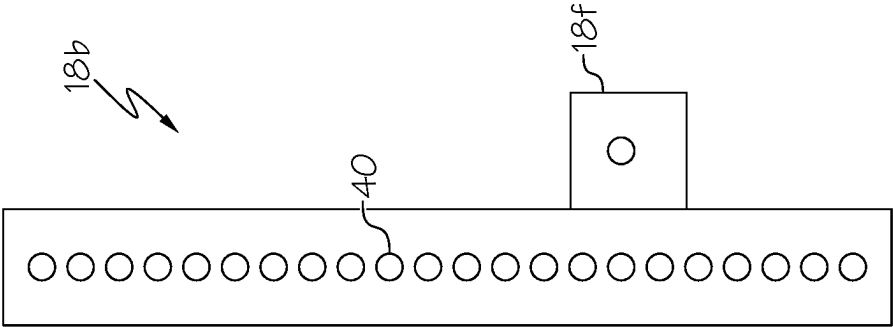


FIG. 13

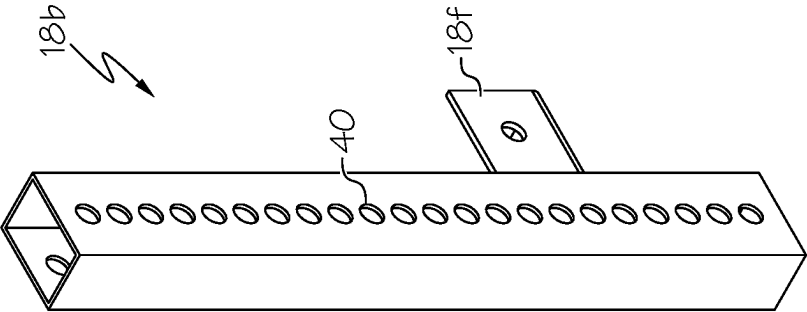


FIG. 12

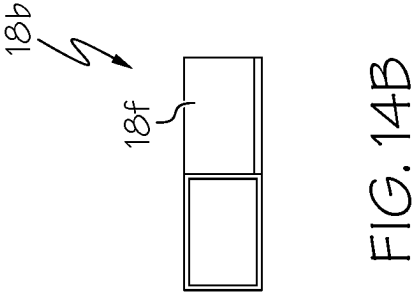


FIG. 14B

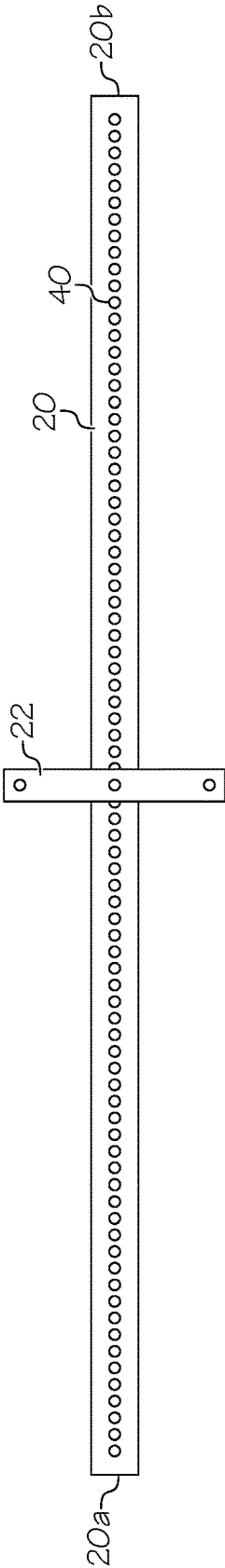


FIG. 16

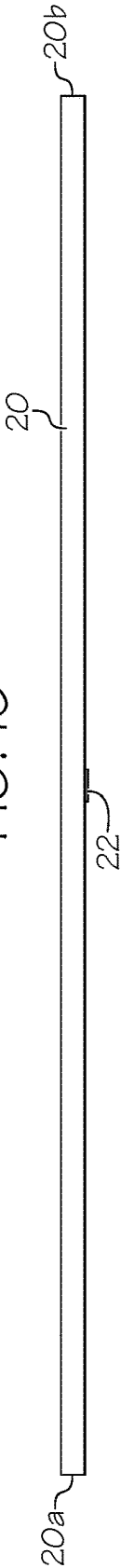


FIG. 17

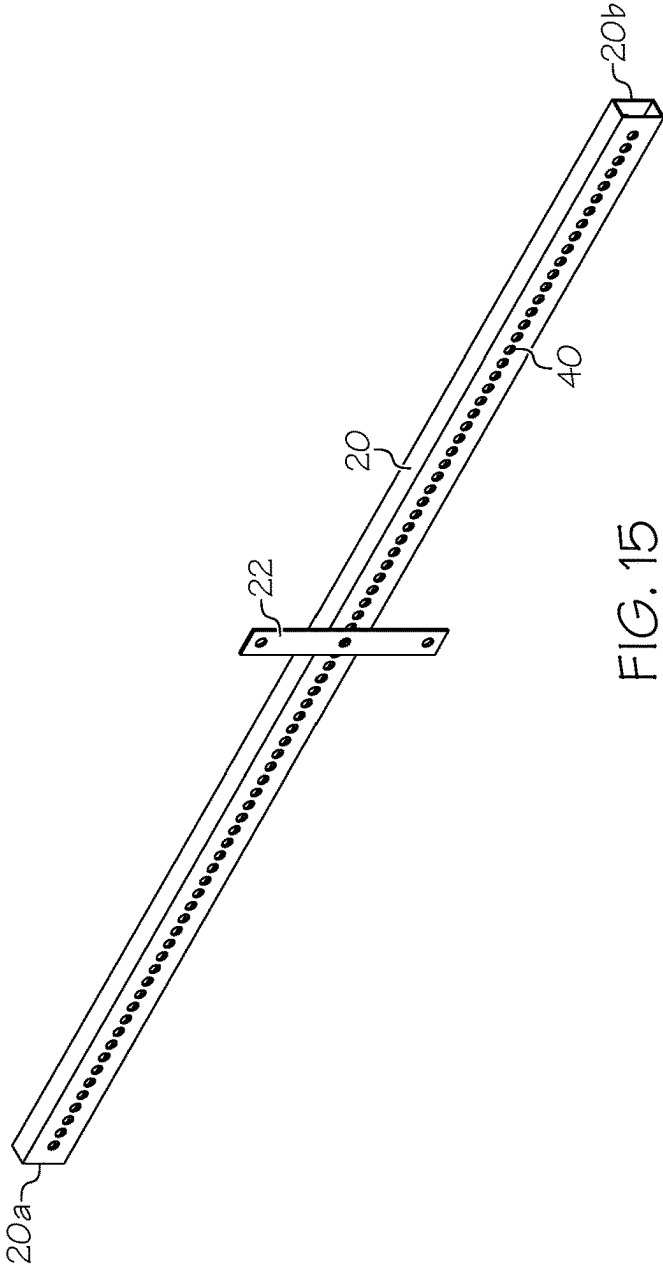


FIG. 15

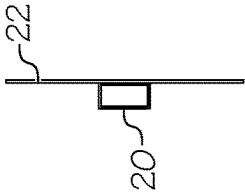


FIG. 18

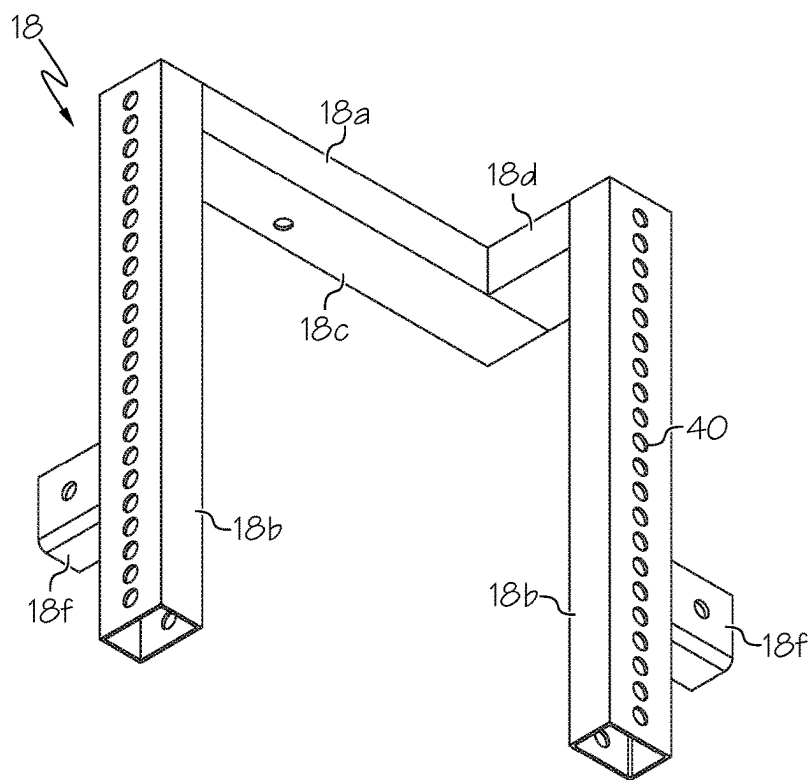


FIG. 19

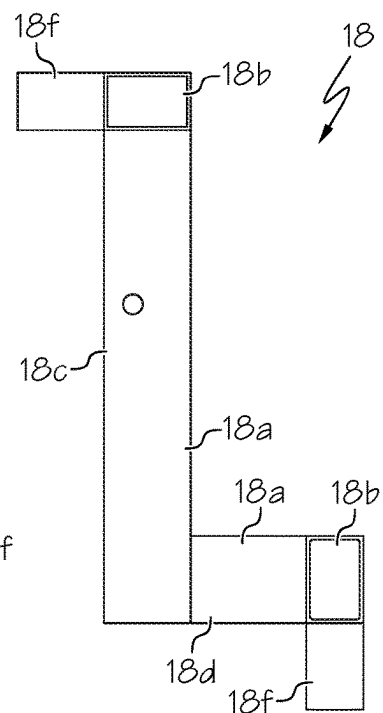


FIG. 20

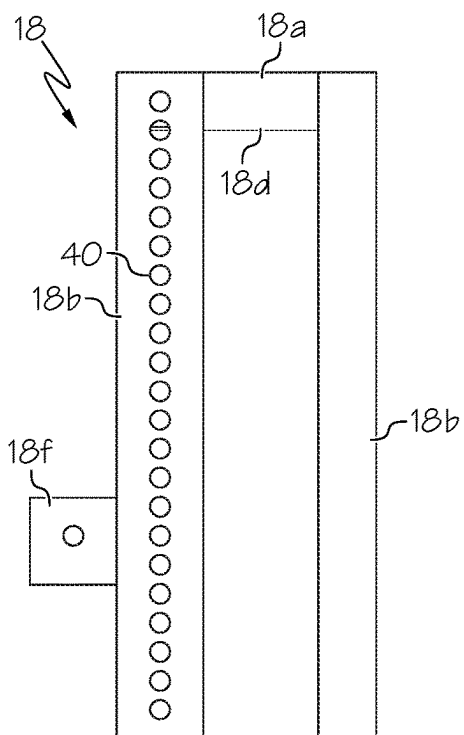


FIG. 21

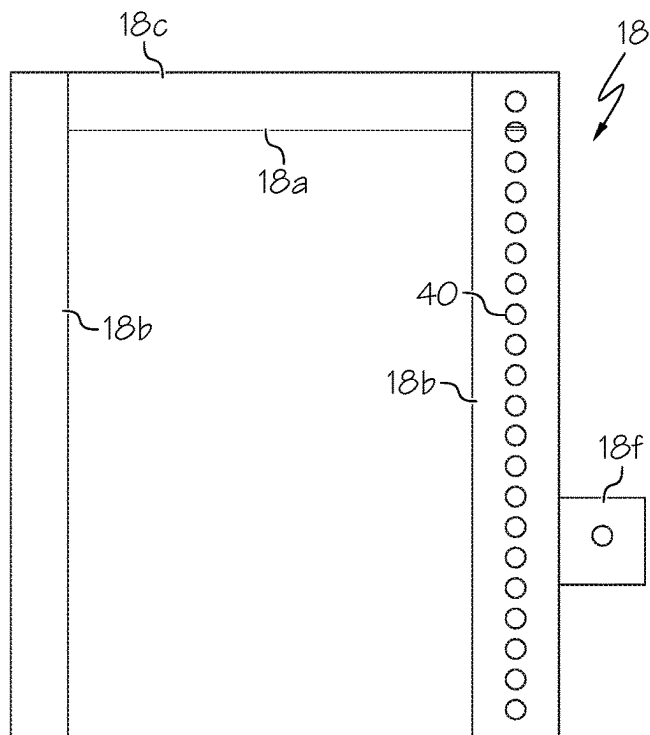


FIG. 22

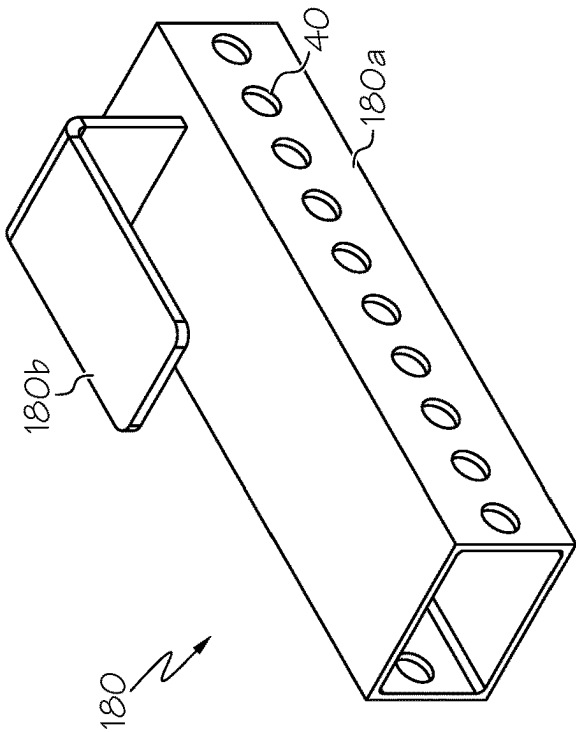


FIG. 23

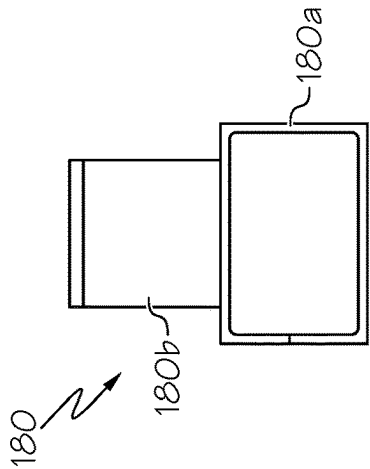
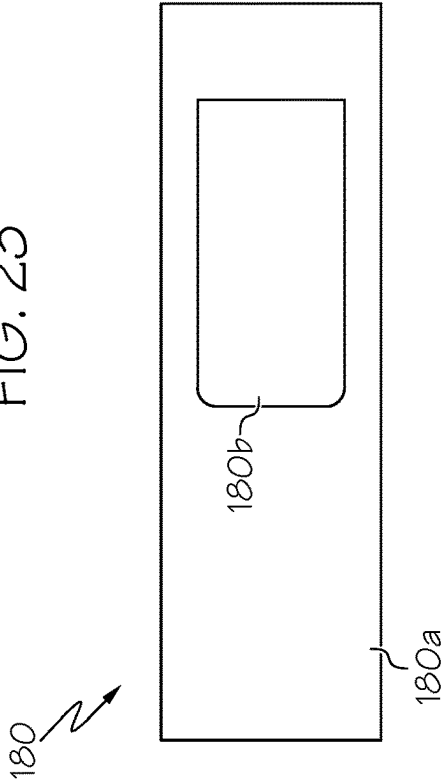


FIG. 24

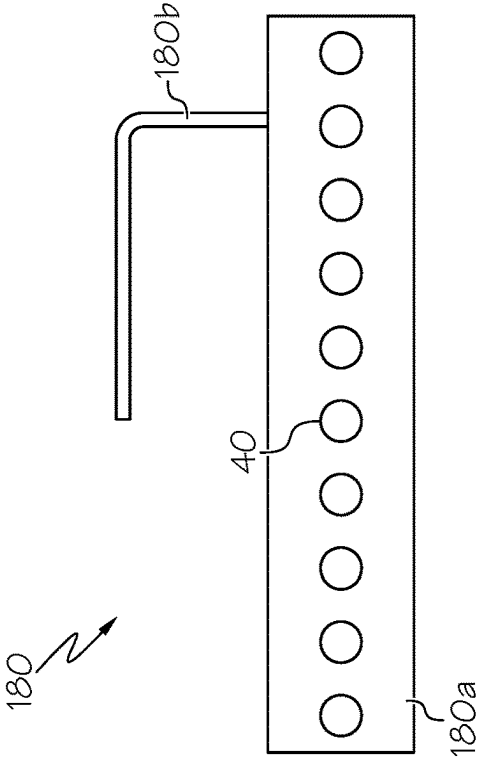


FIG. 26

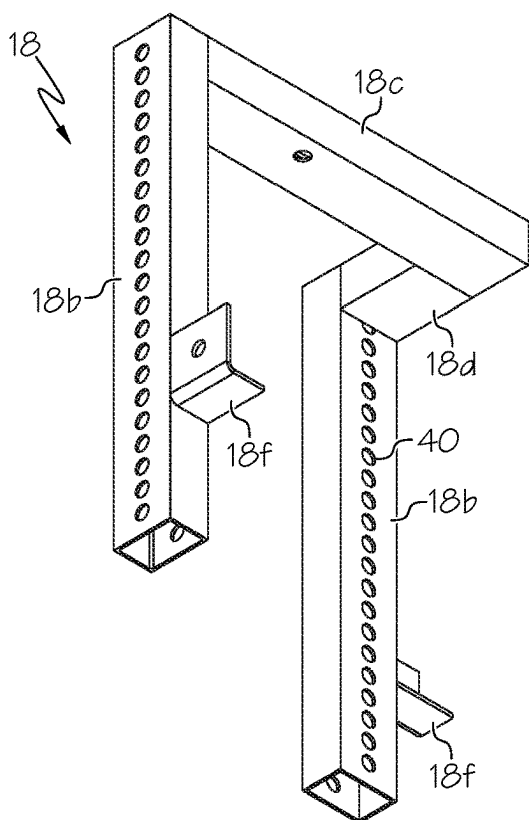


FIG. 27

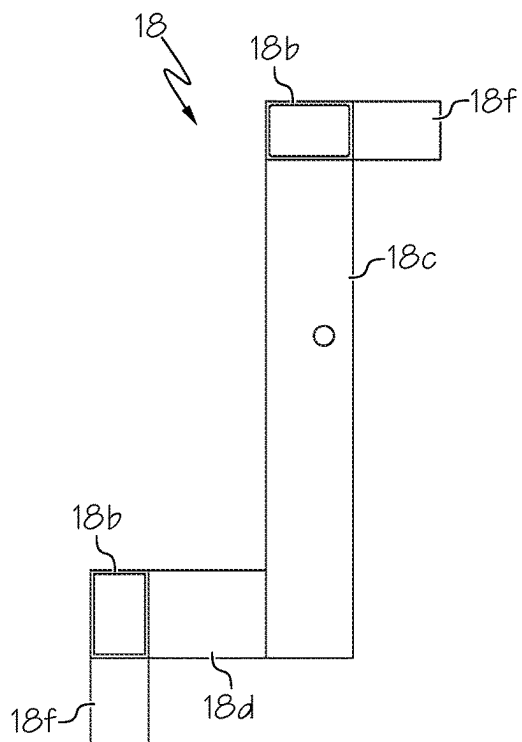


FIG. 28

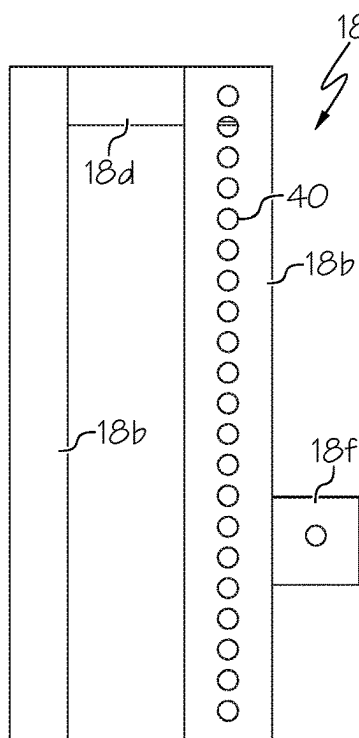


FIG. 29

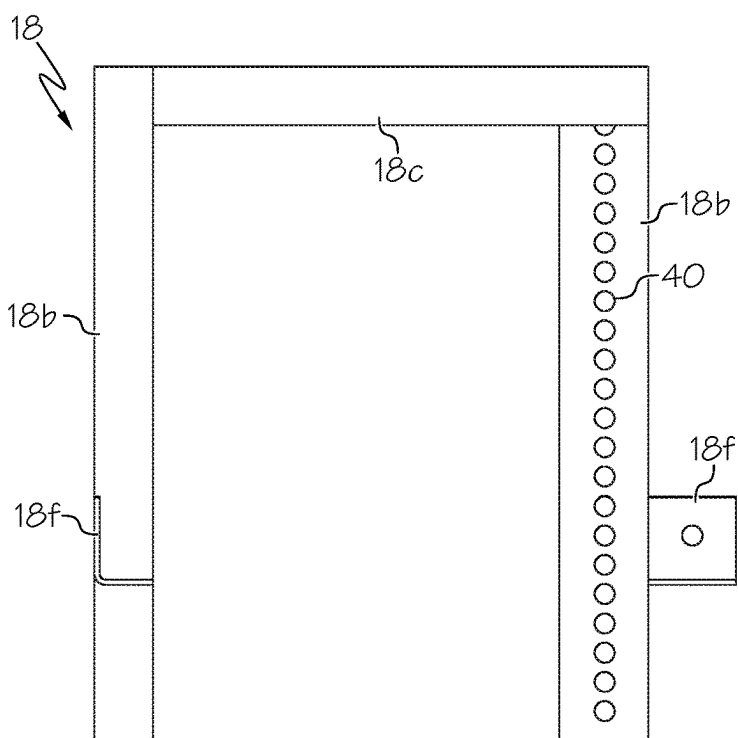


FIG. 30

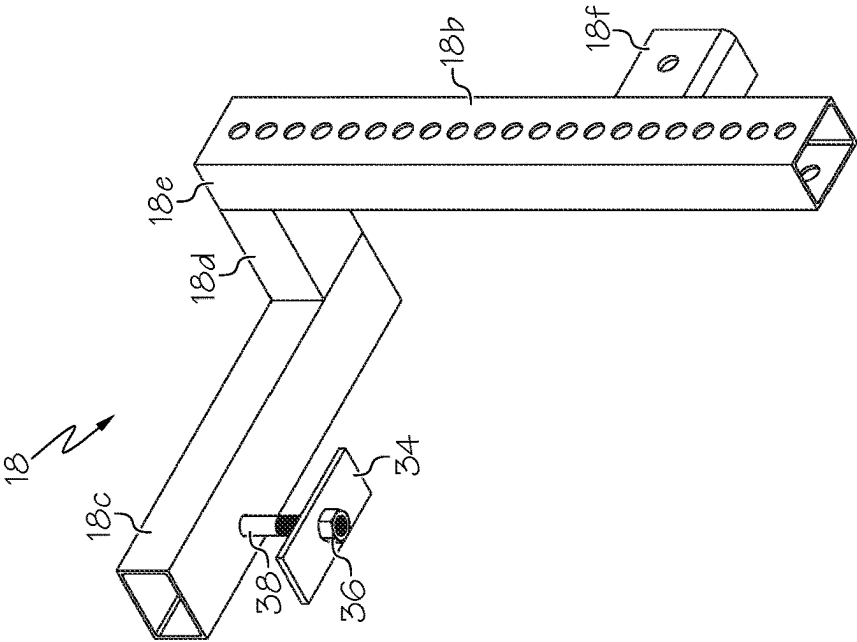


FIG. 31

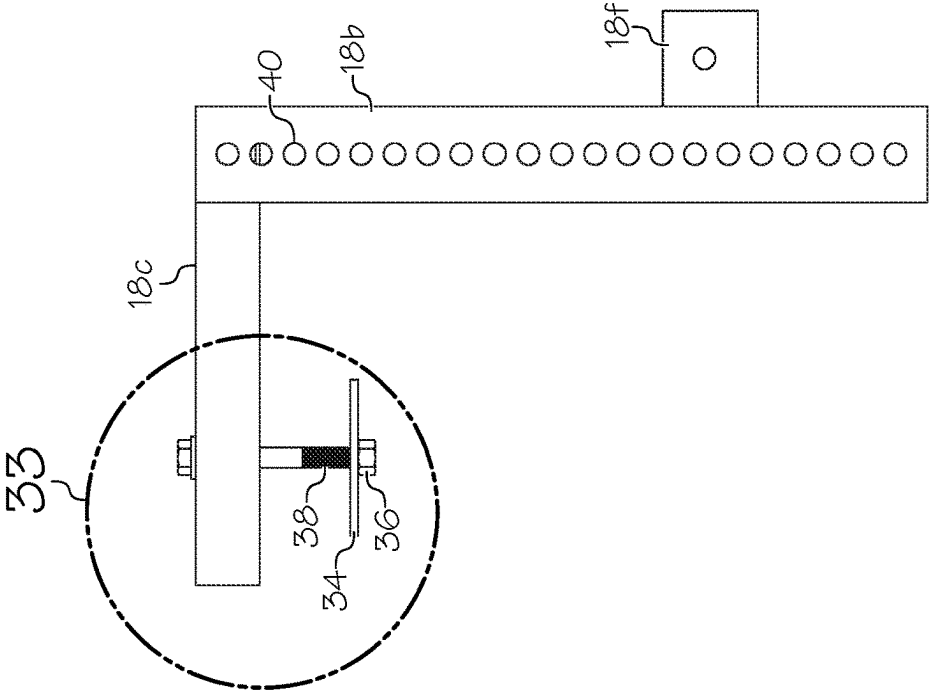


FIG. 32

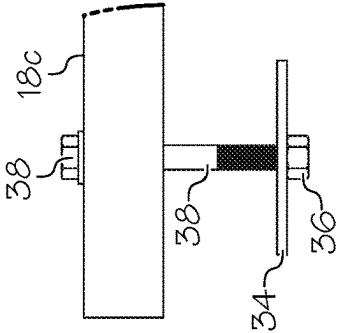


FIG. 33

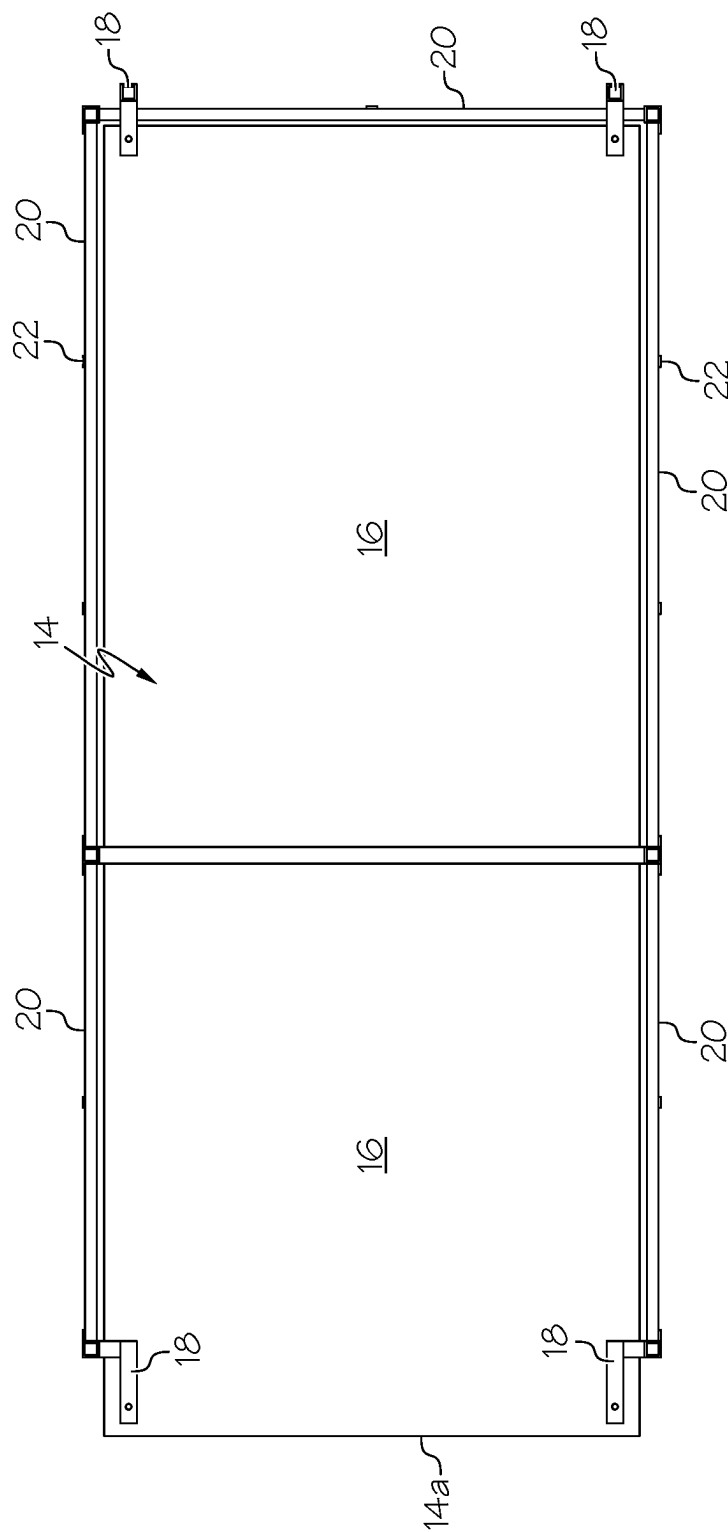


FIG. 34

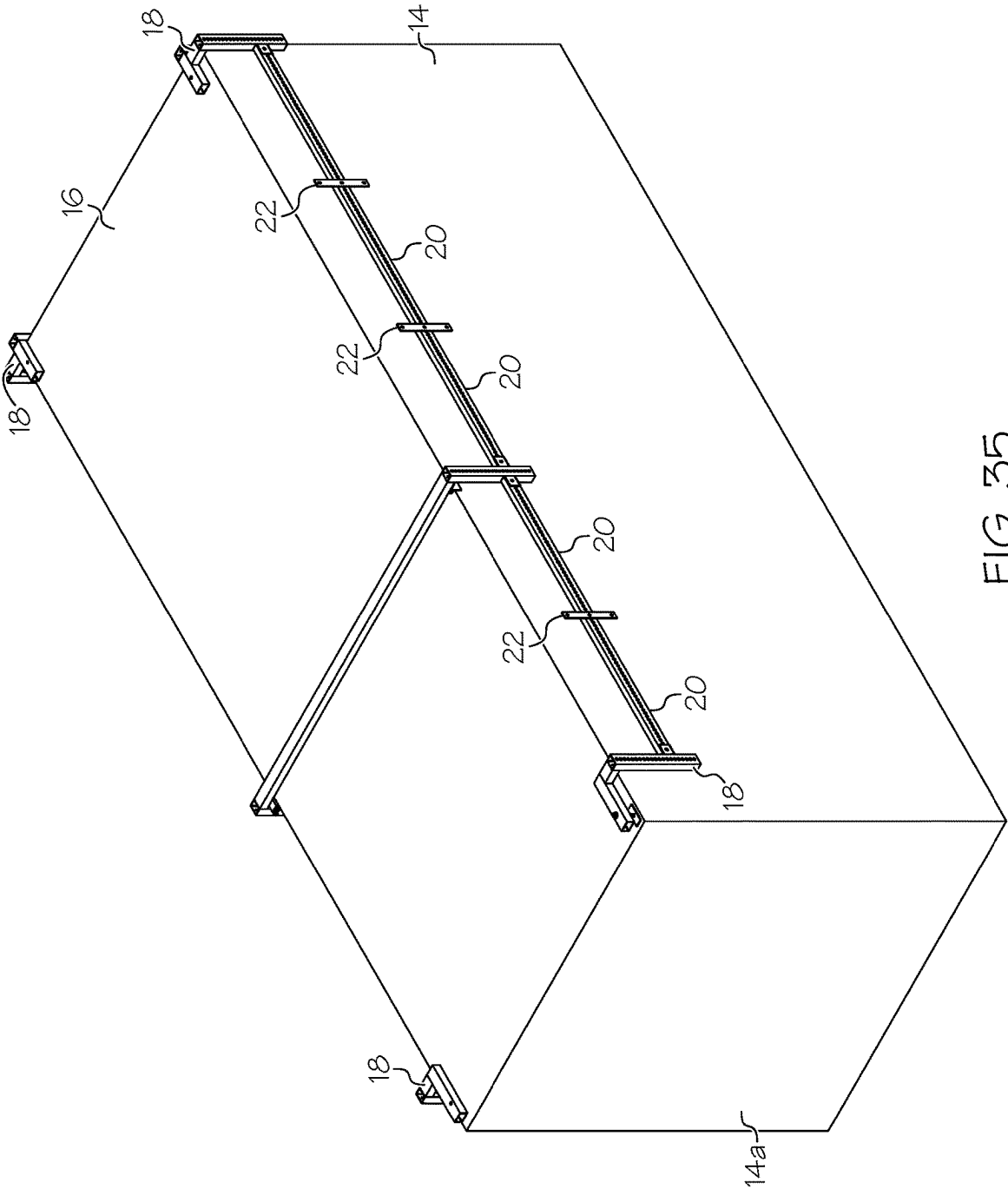


FIG. 35

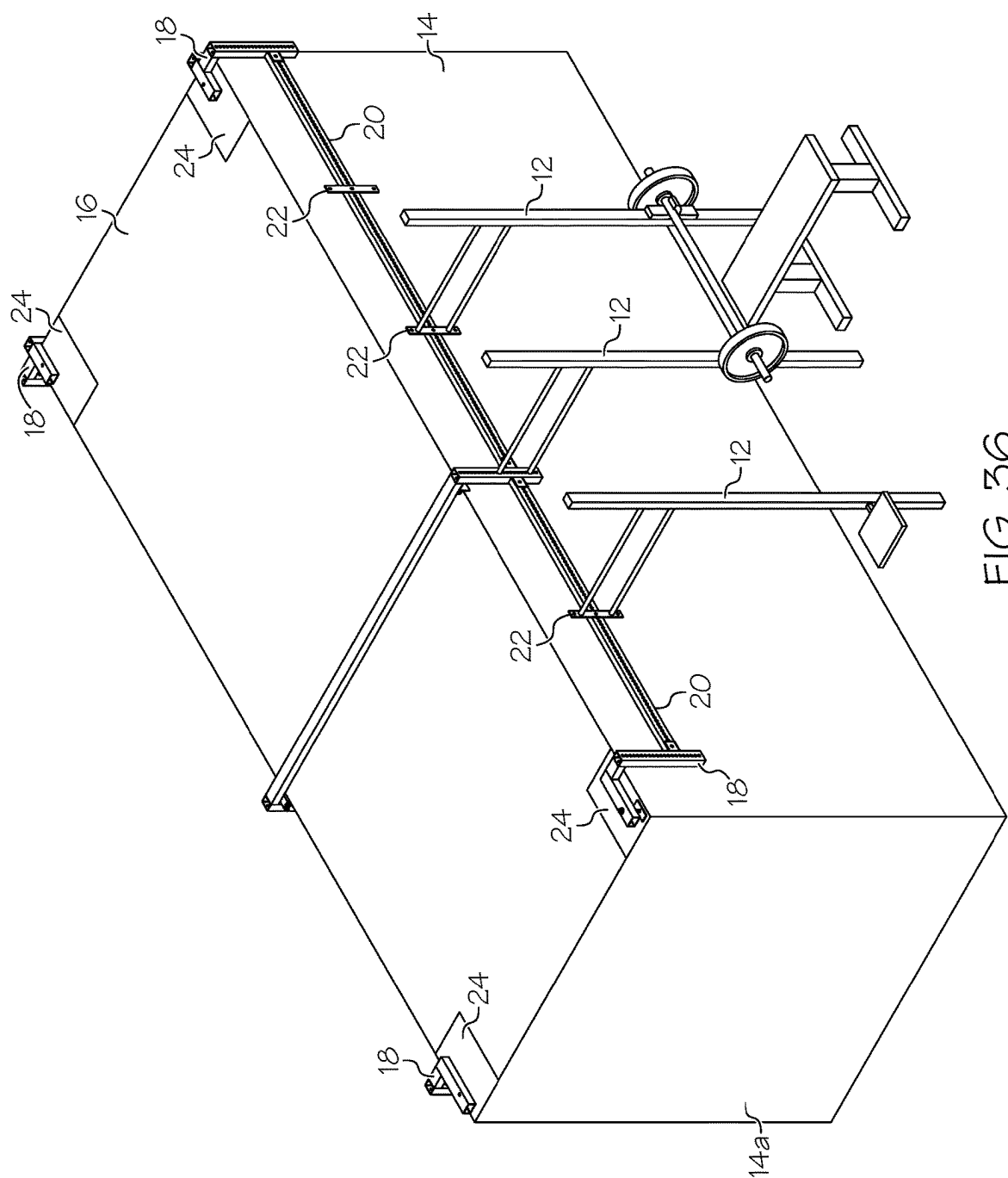


FIG. 36

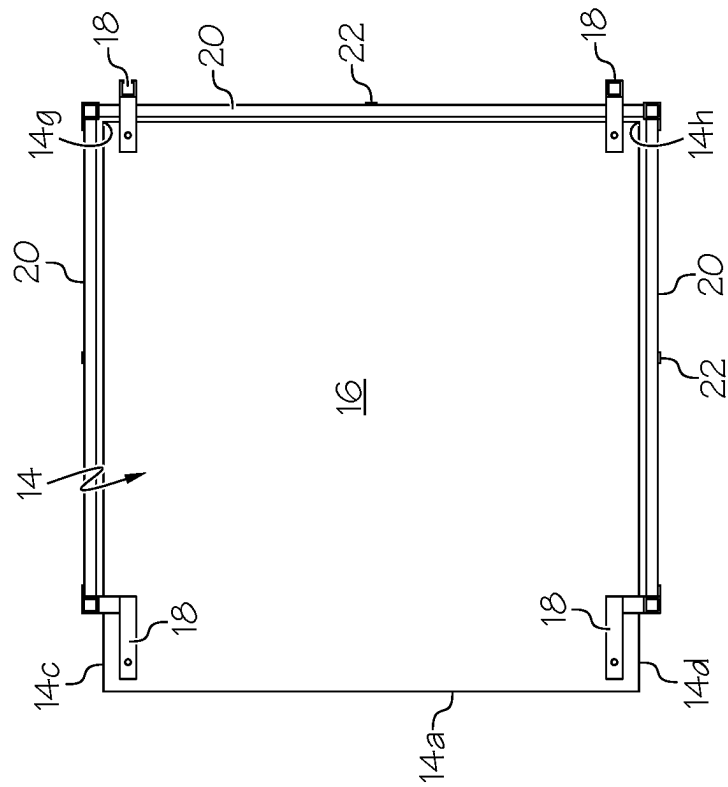


FIG. 37

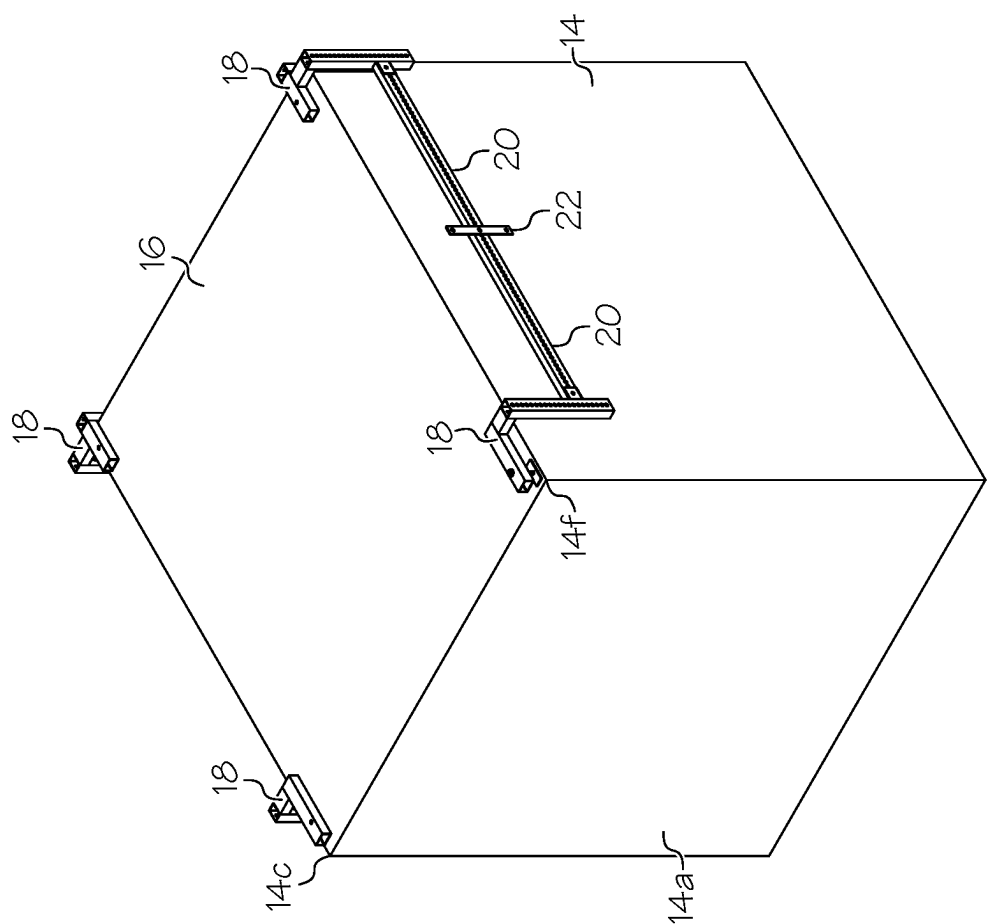


FIG. 38

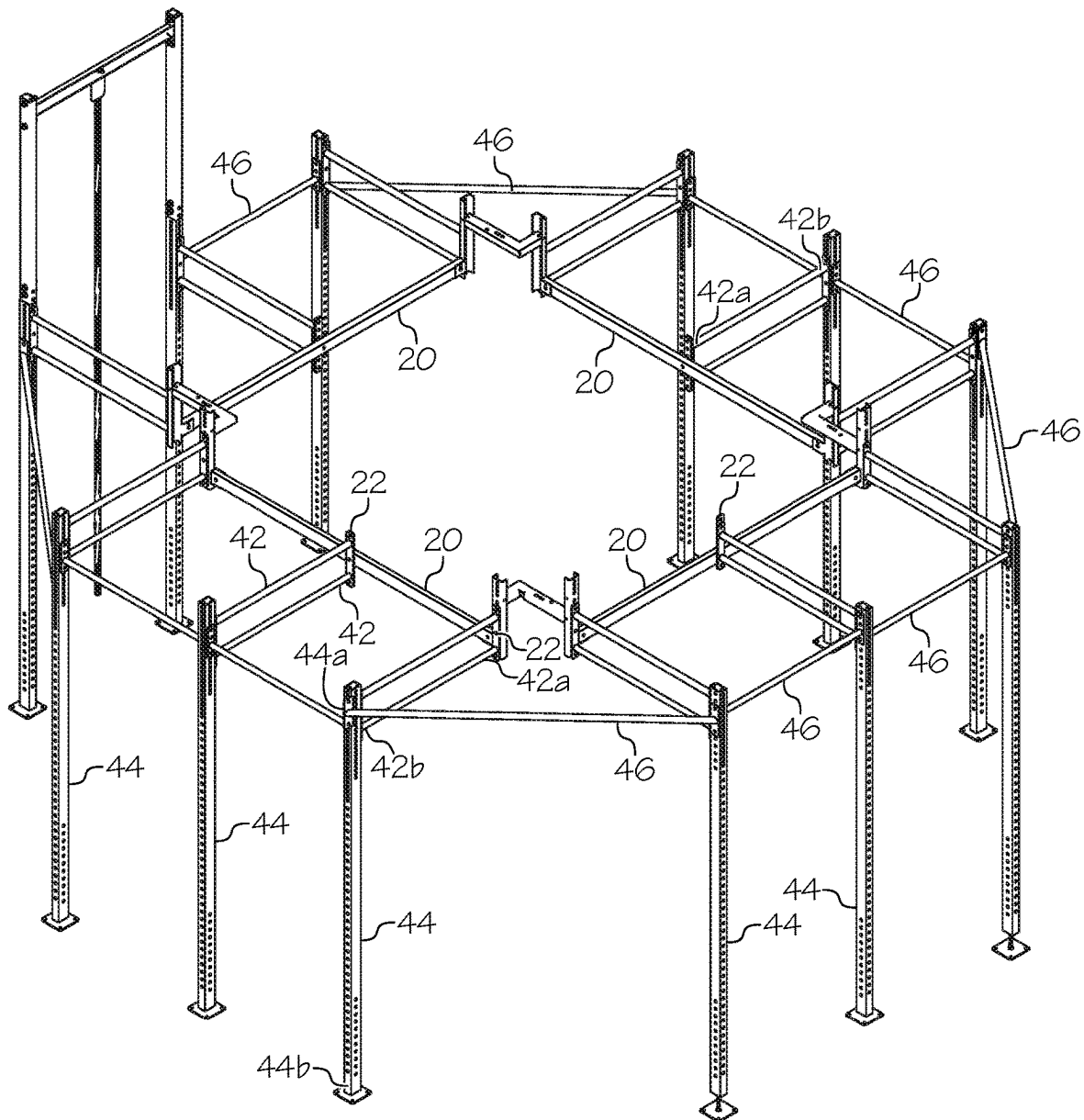


FIG. 39

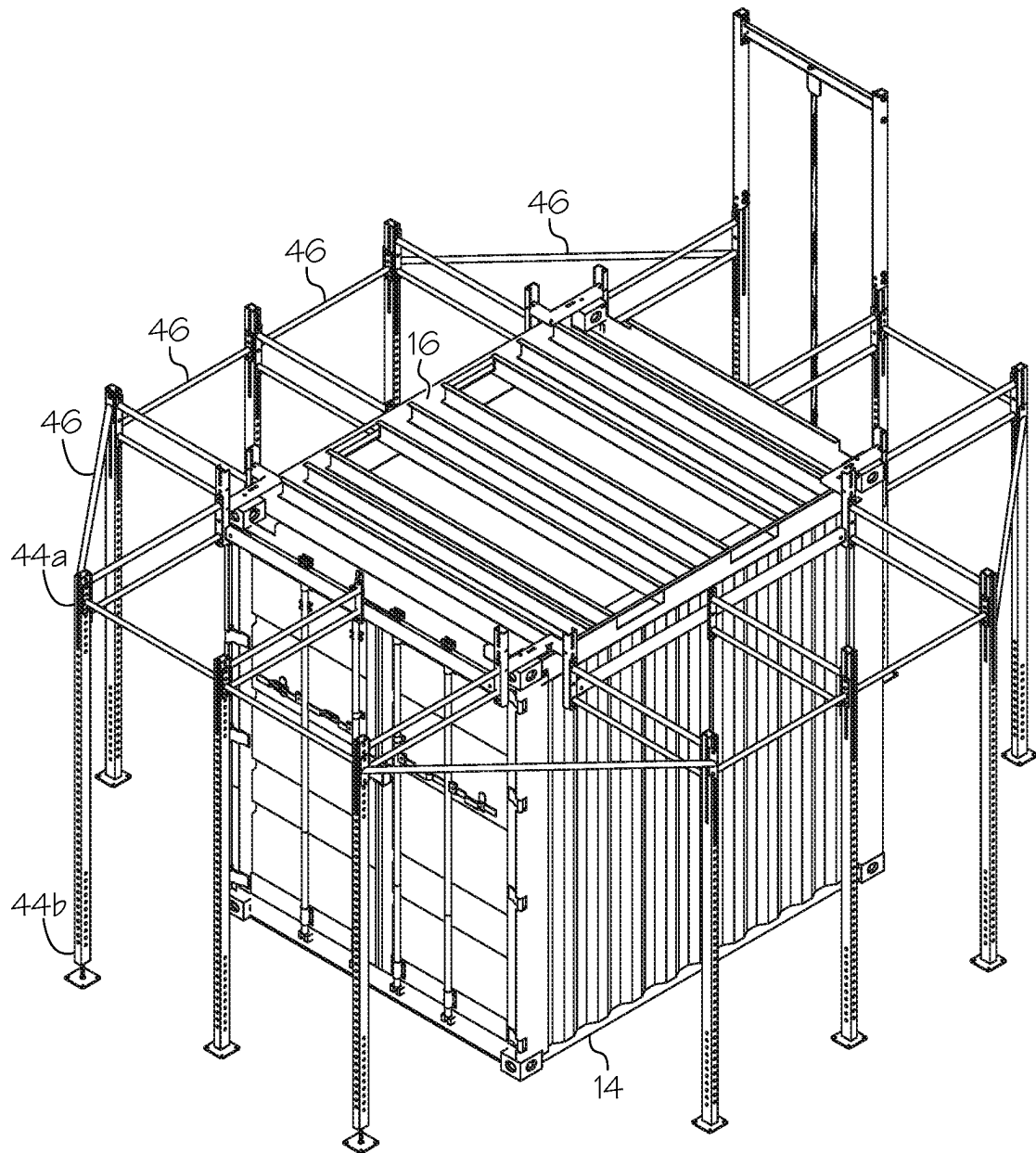


FIG. 40

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ROOFTOP MOUNT FOR EXERCISE EQUIPMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a nonprovisional application of and claims priority from U.S. provisional patent application Ser. No. 62/795,308 filed on Jan. 22, 2019, and is a continuation-in-part application of U.S. nonprovisional patent application Ser. No. 16/749,612 filed on Jan. 22, 2020. The foregoing applications are incorporated in their entirety herein by reference.

FIELD OF THE INVENTION

The invention relates to exercise equipment and gyms. More particularly, the invention relates to a rooftop mount device for connection to a roof of a shipping container for attaching exercise equipment to the device for use as a portable gym.

BACKGROUND

Transport of exercise equipment for use in other locations remote from the home or workplace of the user is both difficult and expensive due to the weight and size of such equipment. Exercisers typically utilize exercise equipment in commercial gyms, home gyms, or gyms located at their workplaces, or when traveling, in hotels. The aforementioned problem inherent in the transportation of this equipment becomes most difficult when the user (and particularly, a group of users) travel to a location that does not have any gym or other exercise facilities. For example, members of the military deployed to remote locations may have no access to local gym facilities. Similarly, attendees at outdoor music concerts and other festivals, and especially those lasting multiple days, may have the desire to continue their usual exercise routines but are unlikely to have access to any exercise equipment or gym facilities.

Conventional examples of portable gyms that use shipping containers for storage and installation of exercise equipment to create mobile “pop-up” gyms install mounting equipment on and through the walls of the containers. By creating new holes in the walls of the containers, the mounting equipment of these conventional portable gyms destroy or harm the integrity of the container by allowing water intrusion into the container as well as animals and debris to enter into the interior spaces of the container. Water intrusion and animal infestations are particularly problematic in that water may corrode or rust both the container as well as exercise equipment stored therein. Animals, such as rodents, may chew and cause damage to the exercise equipment stored in the container or may cause contamination through feces and urine and nesting that can lead to the spread of infectious diseases or of parasites. Rather than maintaining the integrity of the storage container and its contents, such wall-mounted portable gym mounting equipment presents numerous problems that increase the cost of maintenance and shorten the life and usefulness of the container and the equipment stored therein. In the case of military usage, such containers are likely to be shipped via oceangoing ships or may be located after delivery near to the ocean or other bodies of water. In any of these cases, water intrusion and damage through mounting holes drilled

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through the container’s side walls are rife with problems as explained above, and in particular, when exposure to salt-water is likely.

Mounting equipment installed through holes drilled or otherwise made in the side walls of a container can also lead to damage to the container’s wall due to the combined weight of the mounting equipment, exercise equipment installed thereon, and human weight. Although container walls are generally strong, the stress of such weight loads on vertical side walls could result in wall failure, for example, by loosening of the mounting equipment after time with repeated usage so that the attached exercise equipment is unstable, for example, wobbly or otherwise subject to unwanted movement during use. In worst case scenarios, the weight of the foregoing items and person might cause the mounting equipment to pull entirely out of the wall potentially leading to injury of the exerciser or damage to the exercise equipment.

In addition, the standard height of most shipping containers is 8 feet, 8.5 feet, 9 feet, or 9.5 feet. Depending on the type of exercise being performed and on the height of the exerciser, mounting the exercise equipment to the side walls of the container may not allow sufficient clearance above the ground so that the exerciser’s body (e.g., feet or legs) contacts the ground as an exercise is being performed. Wall-mounted portable gym systems provide reduced flexibility in adjusting the height of installation of exercise equipment to the container, as such wall-mount systems are disadvantaged and limited by the height of the container’s walls.

A need exists for a mount device for attaching exercise equipment to a portable gym facility (e.g., a container), which can be packed, stored, and transported easily and inexpensively. A need also exists for a mount device for attaching exercise equipment to a portable gym facility that can be assembled and disassembled easily as the portable gym facility is transported between locations for use by users. A further need exists for a portable gym container into which exercise equipment can be packed and stored when not in use, and to which exercise equipment may be connected for stabilization and support when the equipment is assembled for use by one or more exercisers. Yet another need exists for a mount device that can be used to attach exercise equipment to a portable gym facility without making holes in the wall of the portable gym facility that could allow intrusion of water, animals, or debris. Still another need exists for a mount device that provides vertically adjustable points of attachment to the portable gym facility to permit exercisers to have sufficient available vertical height to perform various exercises using the exercise equipment connected to the mount. Finally, another need exists for a mount device that is installable stably and securely on and to a container without damaging, harming, or destroying the integrity of the shipping container and its walls and that does not present risk of structural failure, damage to equipment, or injury.

SUMMARY

The invention relates to a rooftop mount device for connecting to a roof of a portable gym facility and for attaching exercise equipment thereto. The portable gym facility is a portable container or other portable object having an interior space, such as a shipping container, having a top surface, or roof, to which the rooftop mount device is attachable. The rooftop mount device includes connection members that are attachable to a roof of a

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container. The rooftop mount device can also include one or more horizontal stabilization members. The rooftop mount device can further include equipment attachment elements installed on one or more of the connection members or horizontal stabilization members. The connection members include parts that allow them to be attached to a roof of the container and to allow exercise equipment to be attached to them without contacting a side wall of the container. Exercise equipment is attached to the connection members, horizontal stabilization members, equipment attachment elements, or to a combination of two or more of the foregoing. Each connection member and horizontal stabilization member includes a plurality of apertures across its length for receiving pins, bolts, screws, or other fasteners or attachment apparatuses that are insertable into the apertures to stably and securely connect the exercise equipment to the rooftop mount device. The rooftop mount device provides proper height and stability for the bars, racks, stands, cages, rigs and riggings, and other exercise equipment that are attached to the container's roof so that the equipment does not move during use by an exerciser.

The various items of exercise equipment are storable inside the portable gym (i.e., within the interior space of the container) when not in use and removable from the container when a user desires to exercise. The exercise equipment includes workout bars, racks, stands, rigs and riggings, and cages for calisthenics or bodyweight exercises as well as weights for weight equipment exercises. The workout bars, racks, stands, rigs and riggings, and cages are attachable to the roof of the portable gym container to provide stability and proper height to them when they are used by an exerciser during a workout.

Portable gym systems and kits are also described, which include a rooftop mount device, various items of exercise equipment, and a portable gym. The portable gym system, which can be provided as a kit, is assemblable for use and disassemblable for storage and transport inside the interior space of the container. The portable gym can be transported for use in different locations. The portable gym system can be a kit that can be transported, configured and assembled, and disassembled as desired.

The inventions described provide an advantage in that the rooftop mount device is useful for attaching exercise equipment to a portable gym facility (e.g., a container), which can be packed, stored, and transported easily and inexpensively. The inventions described herein also provide another advantage in that the rooftop mount device is useful for attaching exercise equipment to a portable gym facility that can be assembled and disassembled easily as the portable gym facility is transported between locations for use by users. A further advantage is provided by the inventions described herein in that the rooftop mount device is useful for attaching exercise equipment to a portable gym facility (e.g., to a container) without making holes in the wall of the portable gym facility that could allow intrusion of water, animals, or debris. Still another advantage of the rooftop mount device and other inventions described herein is that they provide vertically adjustable points of attachment to the portable gym facility to permit exercisers to have sufficient available vertical height to perform various exercises using the exercise equipment connected to the mount for ample clearance of the exerciser's body above the substrate on which the container rests. Finally, the rooftop mount device and other inventions described herein provide another advantage in that they are installable stably and securely on and to a container without damaging, harming, or destroying the integrity of the shipping container and its walls and that does

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not present risk of structural failure, damage to equipment, or injury. Rather, the rooftop mount device has an important benefit of being installable on and to a shipping container using existing holes in the roof of the container, for example, the existing holes in the corner castings found at each corner of the roof of a shipping container.

Accordingly, the invention features a rooftop mount device for connecting to a roof of a container and attaching exercise equipment thereto. The rooftop mount device includes at least two connection members that are attachable to a roof of a container and at least one horizontal stabilization member for connecting between two opposing connection members of the at least two connection members when the rooftop mount device is attached to the roof of the container. The exercise equipment is connectable to the rooftop mount device at the connection members, at the horizontal stabilization member, or at both. The exercise equipment is adjustably attachable both horizontally and vertically at a plurality of connection points on the connection members and the horizontal stabilization member.

In another aspect, the invention can feature the exercise equipment being or including workout bars, racks, stands, and cages for calisthenics or bodyweight exercises.

In another aspect, the invention can feature the rooftop mount device having at least four of the connection members.

In another aspect, the invention can feature each connection member connecting to the roof of the container at a corner casting, wherein one corner casting is located at each corner at a top side of the container.

In another aspect, the invention can feature each connection member including a horizontal element and a vertical element. The horizontal element includes an attachment portion at which the connection member is attachable to the roof of the container and an extension portion that extends beyond the horizontal edge of the roof when the rooftop mount device is installed on the roof. The vertical element includes a top portion that connects to the extension portion of the horizontal element and a connection body for attaching the at least one horizontal stabilization member to the vertical element. The vertical element extends downward parallel to, but not in contact with, a side wall of the container.

In another aspect, the invention can feature the horizontal stabilization member including a horizontal brace having a first end and a second end, wherein the first end attaches to at least one of the connection members.

In another aspect, the invention can feature the rooftop mount device further including a center brace having a left vertical portion, a right vertical portion, and a horizontal portion. The left and right vertical portions each attach to the roof of the container at opposing points located at left and right sides of the container between connection members. The horizontal portion extends across the roof and includes a first end that connects to the left vertical portion and a second end that connects to the right vertical portion.

In another aspect, the invention can feature the left vertical portion and the right vertical portion of the center brace each including at least one connection body for attaching the at least one horizontal stabilization member to the vertical portion adjacent to it.

In another aspect, the invention can feature the center brace including a plate for contacting against the roof at one side. The plate extends downward from a bottom of the horizontal portion. The center brace further includes a tension bolt that, when tightened, presses the plate against

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the roof of the container to maintain the at least one horizontal stabilization member in position during use of the exercise equipment.

In another aspect, the invention can feature the at least two connection members each including a vertical portion and a tab protruding from one side. The tab is sized and shaped to fit into an aperture of a corner casting of the container to permit installation of the at least two connection members without tools.

In another aspect, the invention can feature the at least two connection members, the at least one horizontal stabilization member, or both including a plurality of apertures for receiving pins or other fasteners to connect exercise equipment to them.

In another aspect, the invention can feature the rooftop mount device further including one or more equipment attachment elements connected to the at least one horizontal stabilization member, wherein each of the one or more equipment attachment elements is adjustable in its position of attachment to the at least one horizontal stabilization member.

The invention also features a portable gym system that includes an object having a roof and a storage area for holding and storing exercise equipment and a rooftop mount device for connecting to the roof and attaching exercise equipment thereto. The rooftop mount device includes connection members that are attachable to the roof and a horizontal stabilization member for connecting between vertical elements of two opposing connection members when the rooftop mount device is attached to the roof. The exercise equipment is connectable to the rooftop mount device at the connection members, at the horizontal stabilization member, or at both. The exercise equipment is adjustably attachable both horizontally and vertically at a plurality of connection points on the connection members and the horizontal stabilization member.

In another aspect, the invention can feature the portable gym system further including exercise equipment storable in the storage area of the object, wherein at least part of the exercise equipment is attachable to the rooftop mount device.

In another aspect, the invention can feature the object being a shipping container, a storage container, a trailer of a tractor trailer, a box of a box truck, a trailer home, a motor home, a camper, a recreational vehicle, or any other portable object having sufficient height and enclosing an interior space in which exercise equipment is storable.

In another aspect, the invention can feature the system including horizontal stabilization members installed on two, three, or four sides of the roof.

In another aspect, the invention can feature the system including equipment attachment elements attached to at least one horizontal connection member at a first end of each horizontal connection member. The at least one horizontal connection member includes a second end that attaches perpendicularly to an equipment connection member. The equipment connection member includes a top end and a bottom end. At the top end, the equipment connection member is attached to adjacent equipment connection members by horizontal cross members.

In another aspect, the invention can feature the system including one or more rooftop mount devices that connect to all sides of the roof of the object so that the object and its roof are encircled or surrounded by the one or more rooftop mount devices.

The invention also relates to a rooftop mount device for connecting to a roof of a container and attaching exercise

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equipment thereto. The rooftop mount device includes at least two connection members that are attachable to a roof of a container, wherein each connection member includes attachment means for connecting exercise equipment to the connection member without contacting a wall of the container. The exercise equipment is connectable to the rooftop mount device at the connection members and is adjustably attachable and repositionable both horizontally and vertically at a plurality of connection points on the connection members.

A method of the invention can be used for attaching exercise equipment to a roof of a container so that the exercise equipment is secure, stable, and adjustable in its position. The method includes the steps of: (a) attaching at least two connection members to a roof of a container, wherein each connection member includes a vertical element that extends downward parallel to, but not in contact with, a side wall of the container; (b) attaching at least one horizontal stabilization member between the vertical elements of the at least two connection members; and (c) connecting exercise equipment to the at least two connection members, to the at least one horizontal stabilization member, or to both. The exercise equipment is adjustably attachable both horizontally and vertically at a plurality of connection points on the at least two connection members and the at least one horizontal stabilization member.

Another method of the invention can include the at least two connection members each including a vertical portion and a tab protruding from one side, wherein the tab is sized and shaped to fit into an aperture of a corner casting of the container, and wherein the tab is inserted into the aperture of the corner casting to install a connection member without tools.

Unless otherwise defined, all technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. All publications, patent applications, patents and other references mentioned herein are incorporated by reference in their entirety. In the case of conflict, the present specification, including definitions will control.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of one embodiment of a rooftop mount for connecting exercise equipment to a roof of a container.

FIG. 2 is a side elevation view of the rooftop mount device of FIG. 1.

FIG. 3 is a left side elevation view of a center brace of the rooftop mount device of FIG. 1.

FIG. 4 is a front side elevation view of the center brace of FIG. 3.

FIG. 5 is a close-up view of a left end of the center brace of FIG. 3.

FIG. 6 is a close-up view of a right end of the center brace of FIG. 3.

FIG. 7 is a bottom plan view of the center brace of FIG. 3.

FIG. 8 is a perspective view of a left down tube of a connection member of the rooftop mount device of FIG. 1.

FIG. 9 is a front side elevation view of the left down tube of FIG. 8.

FIG. 10 is a left side elevation view of the left down tube of FIG. 8.

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FIG. 11 is a top plan view of the left down tube of FIG. 8.

FIG. 12 is a perspective view of a right down tube of a connection member of the rooftop mount device of FIG. 1.

FIG. 13 is a front side elevation view of the right down tube of FIG. 12.

FIG. 14A is a right side elevation view of the right down tube of FIG. 12.

FIG. 14B is a top plan view of the right down tube of FIG. 12.

FIG. 15 is a perspective view of a horizontal brace of the rooftop mount device of FIG. 1.

FIG. 16 is a front side elevation view of the horizontal brace of FIG. 15.

FIG. 17 is a top plan view of the horizontal brace of FIG. 15.

FIG. 18 is a right side elevation view of the horizontal brace of FIG. 15.

FIG. 19 is a perspective view of a left rear corner connection member of the rooftop mount device of FIG. 1.

FIG. 20 is a bottom plan view of the left rear corner connection member of FIG. 19.

FIG. 21 is a left side elevation view of the left rear corner connection member of FIG. 19.

FIG. 22 is a right side elevation view of the left rear corner connection member of FIG. 19.

FIG. 23 is a perspective view of tab mount device for attachment to a shipping container.

FIG. 24 is a bottom plan view of the tab mount device of FIG. 23.

FIG. 25 is a rear side elevation view of the tab mount device of FIG. 23, wherein the tab mount device is shown oriented horizontally.

FIG. 26 is a left side elevation view of the tab mount device of FIG. 23, view of the tab mount device of FIG. 23, wherein the tab mount device is shown oriented horizontally.

FIG. 27 is a perspective view of a right rear corner connection member of the rooftop mount device of FIG. 1.

FIG. 28 is a bottom plan view of the right rear corner connection member of FIG. 27.

FIG. 29 is a left side elevation view of the right rear corner connection member of FIG. 27.

FIG. 30 is a right side elevation view of the right rear corner connection member of FIG. 27.

FIG. 31 is a perspective view of a right front corner connection member, with bolt and nut plate, of the rooftop mount device of FIG. 1.

FIG. 32 is a right side elevation view of the right front corner connection member of FIG. 31.

FIG. 33 is a close-up view of the bolt and nut plate attached to the right front corner connection member of FIG. 31.

FIG. 34 is a top plan view of the rooftop mount device of FIG. 1 attached to a shipping container.

FIG. 35 is a perspective view of the rooftop mount device of FIG. 1 attached to the shipping container.

FIG. 36 is a perspective view of the rooftop mount device of FIG. 1 attached to the shipping container, along with exercise equipment attached to the rooftop mount device.

FIG. 37 is a top plan view of another embodiment of a rooftop mount device attached to a shipping container that is shorter in length than the shipping container shown in FIG. 1.

FIG. 38 is a perspective view of the rooftop mount device of FIG. 37 attached to the shipping container.

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FIG. 39 is a perspective view of the rooftop mount device of FIG. 39 shown attached to all sides of a shipping container.

FIG. 40 is a perspective view of a rooftop mount device for attachment to all sides of a shipping container, wherein the shipping container is not shown.

DETAILED DESCRIPTION

The present invention is best understood by reference to the detailed drawings and description set forth herein. Embodiments of the invention are discussed below with reference to the drawings; however, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes as the invention extends beyond these limited embodiments. For example, in light of the teachings of the present invention, those skilled in the art will recognize a multiplicity of alternate and suitable approaches, depending upon the needs of the particular application, to implement the functionality of any given detail described herein beyond the particular implementation choices in the following embodiments described and shown. That is, numerous modifications and variations of the invention may exist that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.

The present invention should not be limited to the particular methodology, compounds, materials, manufacturing techniques, uses, and applications, described herein, as these may vary. The terminology used herein is used for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention. As used herein and in the appended claims, the singular forms "a," "an," and "the" include the plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to "an element" is a reference to one or more elements and includes equivalents thereof known to those skilled in the art. Similarly, for another example, a reference to "a step" or "a means" may be a reference to one or more steps or means and may include sub-steps and subservient means.

All conjunctions used herein are to be understood in the most inclusive sense possible. Thus, a group of items linked with the conjunction "and" should not be read as requiring that each and every one of those items be present in the grouping, but rather should be read as "and/or" unless expressly stated otherwise. Similarly, a group of items linked with the conjunction "or" should not be read as requiring mutual exclusivity among that group, but rather should be read as "and/or" unless expressly stated otherwise. Structures described herein are to be understood also to refer to functional equivalents of such structures. Language that may be construed to express approximation should be so understood unless the context clearly dictates otherwise.

Unless otherwise defined, all terms (including technical and scientific terms) are to be given their ordinary and customary meaning to a person of ordinary skill in the art, and are not to be limited to a special or customized meaning unless expressly so defined herein.

Terms and phrases used in this application, and variations thereof, especially in the appended claims, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. As examples of the foregoing, the term "including" should be read to mean "including, without

limitation,” “including but not limited to,” or the like; the term “having” should be interpreted as “having at least”; the term “includes” should be interpreted as “includes but is not limited to”; the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof; and use of terms like “preferably,” “preferred,” “desired,” “desirable,” or “exemplary” and words of similar meaning should not be understood as implying that certain features are critical, essential, or even important to the structure or function of the invention, but instead as merely intended to highlight alternative or additional features that may or may not be utilized in a particular embodiment of the invention.

Those skilled in the art will also understand that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations; however, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C” is used, in general, such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.).

All numbers expressing dimensions, quantities of ingredients, reaction conditions, and so forth used in the specification are to be understood as being modified in all instances by the term “about” unless expressly stated otherwise. Accordingly, unless indicated to the contrary, the numerical parameters set forth herein are approximations that may vary depending upon the desired properties sought to be obtained.

As shown in FIGS. 1 and 2, the invention provides a rooftop mount device 10 (occasionally also referred to herein simply as a “rooftop mount”) for connecting exercise equipment 12 to a roof 16 of a container 14 or other object. For convenience, the roof 16 is described herein with reference to the roof of a shipping container 14, e.g., an ISO shipping container of the type used to transport cargo on ships and trains. However, in other embodiments, the roof 16 may be part of an object 14 such as, for example, a storage container, a trailer of a tractor trailer, a box of a box truck, a trailer home, a motor home, a camper, a recreational vehicle, or any other portable object having sufficient height and enclosing an interior space in which exercise equipment is storable. The container 14 is a shipping container (e.g., an ISO shipping container) in exemplary embodiments, for

example, the type of shipping container that is used to ship goods via oceangoing freight ships. In other embodiments, the container 14 may be another type of storage container. The container 14 includes side walls and a roof 16. The container 14 also includes a floor in most embodiments. In exemplary embodiments, the container 14 includes four vertical side walls that attach to the floor of the container at their bottom edges and to the roof of the container at their top edges. In those embodiments, the shape of the container is a parallelepiped and is either rectangular or square in longitudinal cross-section. In other embodiments, the side walls of the container may be curved or angled rather than vertical. In other embodiments, rather than four side walls, the container may have three, five, six, seven, eight, or more side walls. Although the container is constructed from metal (e.g., steel) in exemplary embodiments, other embodiments of the container may be constructed from a semimetal like aluminum, an alloy, a plastic or other hydrocarbon-based synthetic material, wood, or any other suitable material that is sufficiently strong, durable, and weather-resistant to store and transport the exercise equipment.

A bottom of the container 14 can rest upon a substrate such as, for example, the ground, a floor, a paved surface, a manufactured surface, a platform, a rail car, or a ship deck. Portions of the exercise equipment 12 are also placed upon and in contact with the substrate.

In most embodiments, the container 14 has a roof 16 supported by a front wall 14a, an opposing rear wall 14b, a left wall 14c, and an opposing right wall 14d, which connect to the roof at their top edges. The container 14 further includes a floor connected to the four walls at their bottom edges. Standard sizes of containers with which the rooftop mount device may be used include ISO shipping containers that are 5 feet, 6.5 feet, 8 feet, 10 feet, and 20 feet in length. However, embodiments of the rooftop mount device may be constructed to fit containers having lengths that are different lengths including intermediate lengths (e.g., 6, 7, 9, 11, 12, 12.5, 15, 16 or 17.5 feet), shorter lengths (e.g., 3, 4, or 4.5 feet), and longer lengths (e.g., 21, 22.5, 24, 25, 27.5, 30, 32, 40, 45, 48, 53, and 60 feet) so that containers of any length, including non-standard sized shipping containers, can be fitted with a rooftop mount device.

On its roof 16, the container 14 includes a front left corner 14e, a front right corner 14f, a rear left corner 14g, and a rear right corner 14h. In the case of containers that are rectangular in longitudinal cross-section, most such containers have doors or a door on one of the short sides of the container rather than on the long sides, although containers may be constructed with doors on any side or on multiple sides. An exemplary “long” container 14 (i.e., one that is 20 feet or more in length) is shown in FIGS. 34-36 with a rooftop mount device 10 of appropriate and corresponding length and width installed on its roof 16. An exemplary “short” container 14 (i.e., one that is less than 20 feet in length) is shown in FIGS. 37-38 with a rooftop mount device 10 of appropriate and corresponding length and width installed on its roof 16. In these examples, the container’s door or doors are present on the front side 14a of the container 14 (that is, at the left side of the drawing) in FIGS. 34-36. Likewise, the container’s door or doors are present on the front side 14a of the container 14 (that is, at the left side of the drawing) in FIGS. 37-38 also. The presence of the doors accounts for the lack of a horizontal stabilization member 20 installed in front of that side of each container in those drawings, which could block access to the interior compartment of the containers 14 by preventing the opening of their doors.

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The rooftop mount device **10** includes at least two connection members **18** that are attachable to a roof **16** of a container **14** and a horizontal stabilization member **20**. As shown in exemplary embodiments in FIGS. 1, 34, and 37, the rooftop mount device **10** can include a connection member **18** installed on and to each corner of the roof **16** (thus, four connection members **18** in total) and three horizontal stabilization members **20** (i.e., one installed in front of each of three sides of the container **14**).

In some embodiments, the rooftop mount device **10** may include only a single connection member **18**, which is attached to the roof **16** of the container **14**. In such embodiments, exercise equipment **12** capable of being connected to and supported by only a single connection member **18** is attached to the single connection member. Such embodiments having only a single connection member **18** need not include a horizontal stabilization member **20** or any equipment attachment elements **22**; however, in some embodiments having only a single connect member, one or more equipment attachment elements **22**, one or more horizontal stabilization elements **20**, or a combination of one or more of each are attached to the connection member.

The rooftop mount device **10** can also include one or more equipment attachment elements **22** for connecting exercise equipment **12** to the rooftop mount **10**. The equipment attachment elements **22** are bars or plates having one or more holes or apertures to which exercise equipment **12** may be connected via pins or other fasteners (not shown in the drawings). The equipment attachment elements **22** can be generally rectangular in shape and have a first end and a second end. These equipment attachment elements **22** can be attached to the connection members **18**, but in exemplary embodiments, are attached to the horizontal stabilization members **20**. When attached to a horizontal stabilization member **20**, an equipment attachment element **22** is oriented vertically so that it is installed on a front portion of the horizontal stabilization member in a perpendicular orientation relative to the horizontal stabilization member. Each equipment attachment element **22** may be adjusted vertically and horizontally in its position of attachment to the horizontal stabilization member **20** to which it is attached by removing the pin or other fastener from the aperture in which it is first inserted, moving the equipment attachment element **22** to a new position either or both horizontally and vertically, and inserting the pin or other fastener into a different hole of the equipment attachment element **22** or of the horizontal stabilization member. For example, if the user desired to connect the exercise equipment **12** at a higher or lower height, the pin (or other fastener) can be removed from the equipment attachment element **22** to disconnect the equipment attachment element from the horizontal stabilization member **20** at its point of attachment thereto. The equipment attachment element **22** can then be adjusted lower or higher relative to the horizontal stabilization member **20**. Once a new desired height is reached, the pin (or other fastener) can be inserted into an aperture of the equipment attachment element **22**, and through it, into the horizontal stabilization member **20** to fix the equipment attachment element **22** in position on and to the horizontal stabilization member at the selected, desired height.

Each equipment attachment element **22** can also be adjusted horizontally by a similar process as described in the preceding paragraph. For example, if the user needs to accommodate the connection of an item of exercise equipment that is larger or smaller than allowed by the then-current connection positions of the equipment attachment elements **22**, the pin (or other fastener) can be removed from

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at least one equipment attachment element **22** to disconnect the equipment attachment element from the horizontal stabilization member **20** at its point of attachment thereto. The equipment attachment element **22** can then be adjusted horizontally in position along the length of the horizontal stabilization member **20** to move that equipment attachment element nearer to or farther from another equipment attachment element. Once a new desired position is reached, the pin (or other fastener) can be inserted into an aperture of the equipment attachment element **22**, and through it, into the horizontal stabilization member **20** to fix the equipment attachment element **22** in position on and to the horizontal stabilization member at the selected, desired horizontal position.

Various items of exercise equipment **12** may be attached to the connection members **18**, the horizontal stabilization members **20**, and to the equipment attachment elements **22**. For example, certain items of exercise equipment may be attached in cantilever fashion to one or more components of the rooftop mount device.

One exemplary embodiment of a connection member **18** suitable for connection to a left rear corner **14g** of a container **14** is shown in FIGS. 19-22. A corresponding embodiment of a connection member **18** suitable for connection to a right rear corner **14h** of the container **14** is shown in FIGS. 27-30. In exemplary embodiments, the connection members **18** are attachable to the roof **16** at existing holes in the container **14** such as, for example, the container's corner castings **24**, which are typically present on each top corner of a shipping container's roof **16**. For example, the connection members **18** can be installed at and through holes such as those at which the container's door or doors are installed. In this way, the rooftop mount **10** can be used to attach exercise equipment **12** to the container **14** without making new, additional holes in the container that could allow intrusion of water, animals, and debris. In other embodiments, additional holes may be drilled into and through the roof of the container for installation of the rooftop mount. However, with most containers, and particularly those having corner castings in the corners of their roofs, drilling additional holes in the roof for installation of the rooftop mount device is unnecessary.

The rooftop mount device **10** allows for attachment of exercise equipment **12** at a height of about 9 feet, or approximately the same height as the roof **16** of the container **20**. In some embodiments, the rooftop mount device **10** is installed so that it is flush or near flush with the roof **16** of the container **14** meaning that no part of the rooftop mount extends above (or much above) the height of the container's roof. In other embodiments, vertical elements **18b** of the connection members **18** may be constructed so that, when a connection member is installed on the roof, such vertical element **18b** extends above the height of the roof (e.g., 6 inches, 12 inches, 18 inches, 2 feet, 3 feet, 4 feet, 5 feet, 10 feet, or more above the roof) to accommodate and allow for connection of exercise equipment that is taller than the roof and in other cases where attachment of the exercise equipment at heights greater than the height of the container is desired. Attachment to the roof permits optimal height for performing exercises using the exercise equipment. For example, if the exerciser is 6 feet tall or taller, the rooftop mount permits exercise equipment, such as a pull-up bar, to be installed at about 9 feet or higher from ground level, which allows a sufficient distance for the exerciser to raise and lower his or her body without touching the ground. Similar devices that attach to a side wall of a container permit installation of such bars at a maximum height of

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about 7.5 feet above ground level, which provides insufficient distance for the exerciser to perform pull-ups and other exercises without the exerciser's feet or legs touching the ground.

Although the rooftop mount device **10** is attached to the roof **16** of a shipping container **14** in exemplary embodiments, in other embodiments, the rooftop mount **10** may be attached to a roof **16** of another object **14**, e.g., to the roof of an automobile, an RV, a camper, a mobile home, a trailer, or any other object of suitable height for attachment of the rooftop mount to which exercise equipment can be connected.

As shown in FIG. 36, the exercise equipment **12** includes bodyweight workout racks and other exercise equipment that are mountable to the roof **16** of the portable container **14** via connection to the rooftop mount **10**. Such workout racks **12** include, for example, workout bars, racks, stands, rigs and riggings, and cages for calisthenics or bodyweight exercises. Climbing ropes may also be attached to the rooftop mount device. The portable gym system can include one or multiple pieces of exercise equipment. The exercise equipment includes one or more top horizontal components. Each top horizontal component features a connector capable of attaching to a connection member of the rooftop mount. Each piece of exercise equipment also includes one or more (and generally, at least two) feet that contact a substrate (e.g., the ground or a floor or paved area) to support the exercise equipment on a side that is away from the portable container. One or more of the feet may be height-adjustable feet. For example, an upright rack can include adjustable feet that allows the exerciser to increase or decrease the height of that piece of exercise equipment.

As shown in the drawings, the rooftop mount device **10** includes at least two connection members **18** that are attachable to the roof **16** of the portable container **14**. The rooftop mount device **10** also includes at least one horizontal stabilization member **20**, and in exemplary embodiments, at least two equipment attachment elements **22** connected to each horizontal stabilization member. In exemplary embodiments, the system includes four connection members with one connection member being installed at each corner of the container's roof. In other embodiments, the system can include one, two, three, five, six, seven, eight, or more connection members. In embodiments of the system that do not include connection members being installed in each corner of the container roof, the connection members may each be attached to the roof of the portable container on the same side of the roof. The connection members are connected to the roof by bolts, although other fasteners like screws, nails, rivets, brackets, hooks, or any other suitable attachment means for connecting the rooftop mount to the roof of the portable container. The connection members are removably attached to the portable container's roof, although in some embodiments, they may be permanently attached.

In certain instances, when a small container **14** is used (e.g., a container that is 5, 6.5, or 8 feet in length), connection members **18** may be attached to the roof **16** of such container and exercise equipment **12** may be connected only to the connection members. In such embodiments that include small containers, horizontal stabilization members **20** may be excluded during assembly and installation of the rooftop mount device to the roof.

In some embodiments, one or more of the connection members **18** may be attached to a different side or sides of the roof **16** than the side of the roof to which at least one of the connection members is attached. For example, in an

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embodiment of the system in which the portable container **14** is shaped as a parallelepiped and that includes more than two connection members **18**, the first and last connection members **18** may be attached each to the roof **16** at opposing sides of the portable container **14** while the other connection members between them are attached to a side of the roof that connects to each of the opposing sides. Each connection member **18** may be connected to the roof **16** of the container **14** at a corner casting **24**, which include premade holes that can be utilized to attach the rooftop mount device to the storage container. Most storage containers **14** include a corner casting **24** at each corner at a top side of the container, i.e., on the roof **16**.

In some embodiments, each connection member **18** includes at least one horizontal element **18a** and at least one vertical element **18b** as shown in FIGS. 19-22 and 27-30. The horizontal element **18a** of the connection member **18** includes an attachment portion **18c** at which the connection member is attachable to the roof **16** of the container **14** and an extension portion **18d** that extends beyond the horizontal edge of the roof **16** when the rooftop mount device **10** is installed on the roof. The extension portion **18d** of the horizontal element **18a** allows attachment of the horizontal stabilization member **20** and exercise equipment **22** to the vertical element **18b** at a distance sufficiently far enough away from the container's side wall to avoid the wall impeding free range of motion in performing exercises. The vertical element **18b** of the connection member **18** includes a top portion **18e** that connects to the extension portion **18d** of the horizontal element **18a** and a connection body **18f** for attaching the at least one horizontal stabilization member **20** to the vertical element. The vertical element **18b** extends downward parallel to, but not in contact with, a side wall of the container **14**.

Each vertical element **18b** of a connection member **18** may include at least one connection body **18f** for receiving and holding one end of a horizontal stabilization member **20**. The connection body **18f** can be a bracket (as shown in FIGS. 19-22 and 27-30) attached to or constructed as part of the vertical element **18b** of the connection member **18**. The connection body **18f** can also be a ledge, a plate, or any other suitable structure capable of receiving, holding, and retaining one end of the horizontal stabilization member **20** in position. A pin or other fastener (not shown in the drawings) may be employed to further secure the end of the horizontal stabilization member **20** to the connection body **18f** of the connection member's vertical element **18b**. In this manner, a first end of the horizontal stabilization member **20** is secured to and supported by a vertical element **18b** of a first connection member **18** while the horizontal stabilization member's second end is secured to and supported by a vertical element **18b** of a second connection member **18**.

FIGS. 8-11 show a vertical element **18b** of a connection member **18** on which the connection body **18f** is located on a left side of the vertical element for attachment of one end (e.g., a second end **20b**) of a horizontal stabilization member **20** to that left side of the vertical element. FIGS. 12-14B show a vertical element **18b** of a connection member **18** on which the connection body **18f** is located on a right side of the vertical element for attachment of one end (e.g., a first end **20a**) of a horizontal stabilization member **20** to that right side of the vertical element. In exemplary embodiments, the horizontal element **18a** and vertical element **18b** of a connection member are permanently connected to form a single, unitary piece. However, in other embodiments, the vertical member **18b** may be removably connectable to the horizontal member **18a** of a connection member **18** so that the

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connection member is formed by the attachment of the horizontal element and vertical element together during assembly.

As shown in FIGS. 1-2 and 34-38, the shape and structure of the front connection members 18 may differ from the shape and structure of the rear connection members 18. For example, front connection members may be generally L-shaped on their top horizontal elements with each having only a single vertical element as shown in FIG. 31, whereas the rear connection members may also be generally L-shaped or T-shaped on their top horizontal elements but with each having two vertical elements as shown in FIGS. 19-22 and 27-30.

As shown in FIGS. 31-33, 35-36, and 38, the connection members installed at the front corners of a container may each include a bolt and nut plate to provide additional stability to the rooftop mount device 10 after installation on a roof 16 of a container 14 so that the rooftop mount does not move when the exercise equipment 12 is in use. The bolt and nut plate includes a bolt 38 that is passed through an aperture on that attachment portion 18c of the horizontal element 18a of a connection member 18. The aperture on the attachment portion 18c passes from the top of attachment portion through its bottom surface. When the bolt 38 is inserted therethrough, a plate 34 having an aperture passing through it is inserted over a free end of the bolt and a nut 36 is inserted over the free end of the bolt to tighten the plate against, for example, an inner surface of a corner casting 24 to secure the connection member to the roof 16 at that corner casting. The bolt and nut plate might also be used to level the rooftop mount device 12 in some circumstances.

In other embodiments as shown in FIGS. 23-26, the at least two connection members 180 can each include a vertical portion 180a and a tab 180b protruding from one side. The tab 180b is sized and shaped to fit into an aperture on top of a corner casting 24 of the container 14 to permit installation of the at least two connection members 180 without tools. The corner castings 24 are located on the roof 16 of the container 14 as part of many containers' standard structural elements.

As shown in FIGS. 15-18, the horizontal stabilization member 20 is, or includes, a horizontal brace having a first end 20a and a second end 20b. The first end 20a of the horizontal stabilization member 20 attaches to a connection member 18. The second end 20b of the horizontal stabilization member 20 attaches to a connection member 18 that is different from the connection member to which the first end 20a was attached. As shown in FIGS. 1, 2, 35, 36, and 38, one or more equipment attachment elements 22 may be connected to the horizontal stabilization members 20. The one or more equipment attachment elements 22 are attachable to the horizontal stabilization member 20 via removable pins or other removable fasteners (not shown in the drawings) as described elsewhere herein, which may be moved from one position to another on the horizontal stabilization member to adjust horizontally the position of connected exercise equipment.

In embodiments of the rooftop mount device 10 installed on containers having lengths of 20 feet or longer, the rooftop mount device can further include a center brace 26 (e.g., a center pinch brace or center pressure brace) as shown in FIGS. 1-7 and 34-36. The center brace 26 has a left vertical portion 26a, a right vertical portion 26b, and a horizontal portion 26c. The left and right vertical portions 26a, 26b each attach to the roof 16 of the container 14 at opposing points located at left and right sides of the container between connection members 18 as shown in FIGS. 1 and 34-36. The

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horizontal portion 26c extends across the roof 16 and includes a first end that connects to the left vertical portion 26a and a second end that connects to the right vertical portion 26b.

The left vertical portion 26a and the right vertical portion 26b of the center brace 26 each include at least one connection body 26d for attaching the at least one horizontal stabilization member 20 to the vertical portion adjacent to it. As shown in FIGS. 4 and 5, the center brace 26 can include a plate 28 for contacting against the roof 16 at one side. The plate 28 extends downward from a bottom of the horizontal portion 26c. As is also shown in those drawings, the center brace 26 further includes a tension bolt 30 that, when tightened, presses the plate 28 against the roof 16 of the container 14 to maintain the at least one horizontal stabilization member 20, which is connected to the center brace, in position during use of the exercise equipment 12. Such pressure or tension can be exerted where the plate 28 makes contact with and against an inner or outer surface of a ridge cap that extends around the roof 16 of the container 14 above the side walls of the container. As shown in the FIGS. 4 and 6, the other end of the center brace 26, opposite to the end at which the plate 28 and tension bolt 30 are located, can also include a plate 32 extending downward from its bottom, which is useful for making contact against an inner or outer surface of a ridge cap that extends around the roof of the container above the side walls of the container.

The at least two connection members 18 include a plurality of apertures 40 for receiving pins or other fasteners to connect exercise equipment 12 to them. These apertures 40 are generally on the front or side surfaces (or both surfaces) of the vertical elements 18b of each connection member 18 as shown in FIGS. 19-22 and 27-30. The spacing between these apertures 40 can be about one inch, although both regular and irregular spacing of shorter and longer lengths between apertures are also contemplated for use with the various embodiments of the rooftop mount device 10, systems, and kits described herein.

The at least one horizontal stabilization member 20 includes a plurality of apertures 40 for receiving pins or other fasteners to connect exercise equipment 12 or equipment attachment elements 22 to it. These apertures 40 are generally on a front surface of each horizontal stabilization member 20 as shown in FIGS. 15-16, but may also be present on other surfaces of the horizontal stabilization member. The spacing between these apertures 40 can be about one inch, although both regular and irregular spacing of shorter and longer lengths between apertures are also contemplated for use with the various embodiments of the rooftop mount device 10, systems, and kits described herein.

The equipment attachment elements 22, which are attached in a generally vertical orientation relative to the horizontal stabilization member 20 to which they are attached, also include a plurality of apertures 40 for receiving pins or other fasteners to connect exercise equipment 12 to them. These apertures 40 are generally on a front surface of each equipment attachment element 22 as shown in FIGS. 1-2, but may also be present on other surfaces of the equipment attachment element. The spacing between these apertures 40 can be about one inch, although both regular and irregular spacing of shorter and longer lengths between apertures are also contemplated for use with the various embodiments of the rooftop mount device 10, systems, and kits described herein.

The rooftop mount device 10 can feature one or more equipment attachment elements 22 connected to the at least one horizontal stabilization member 20, wherein each of the

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one or more equipment attachment elements is adjustable in its position of attachment to the at least one horizontal stabilization member.

The connections between the various parts of the rooftop mount device **10** may be permanent or removable. Each connection member **18** and horizontal stabilization member **20** is hollow, and as described above, includes one or more apertures or holes **40** that extend through an exterior surface and into a hollow cavity for receiving bolts, screws, or other fasteners to attach top horizontal components of the exercise equipment **12**. The top horizontal components of the bars, racks, stands, rigs and riggings, and cages for calisthenics or bodyweight exercises are attachable to the equipment attachment elements of the rooftop mount **10** so as to maintain the exercise equipment **12** in position for use by an exerciser. One or more equipment attachment elements **22** may be connected, permanently or removably, to each horizontal stabilization member **20** in some embodiments. The one or more holes **40** of the equipment attachment elements **22** can be spaced at even intervals or at irregular intervals across a front side of each equipment attachment element. The holes **40** allow the top horizontal components of the exercise equipment **12** to be adjusted vertically for optimal height for a particular exercise or for a particular exerciser (e.g., based on the exerciser's personal height).

In some alternate embodiments, the connection members **18** and one or more of the attached equipment attachment elements **22** may form a single unitary piece that is connected to the portable container's roof **16**. For example, a single combined and unitary connection member and equipment attachment element can be a L-shaped mount, wherein a horizontal arm of the L-shaped mount attaches to the roof of the container, and a vertical arm of the L-shaped mount extends perpendicularly downward from one end of the horizontal arm. The horizontal arm can include a first end that is on top of the container's roof and a second end that overhangs the roof and from which the vertical arm extends downward at a distance from a side wall of the container. The horizontal stabilization member can be attached to the vertical arms of first and second L-shaped mounts. Top horizontal components of the exercise equipment can also be attached to the vertical arms.

In some embodiments, the horizontal stabilization member **20** also may be permanently attached to the equipment attachment elements **22** so as to also form part of the single unitary piece that is connected to the portable container's roof **16**.

The horizontal stabilization member **20** is a generally horizontal bar that is connected (removably or permanently, but preferably removably) at a first end **20a** to a connection member **18** of the rooftop mount **10** and at a second end **20b** to another connection member **18**. For example, the rooftop mount **10** can include a first connection member **18** installed at a left corner of the roof **16** on one side of the container **14** and a second connection member **18** installed at a right corner of the roof on that same side of the container. As shown in FIGS. 34-38, a first end **20a** of the horizontal stabilization member **20** can be attached to the first connection member **18** and a second end **20b** of the horizontal stabilization member **20** can be attached to the second connection member **18**. The horizontal stabilization member **20** may also include one or more holes **40** that permit additional equipment attachment elements **22** to be attached to it at a position or positions desired by the exerciser along the length of the horizontal stabilization member. This adjustability feature allows the exerciser to attach the exer-

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cise equipment (e.g., the top horizontal components of a power rack) to any point on the horizontal stabilization member.

Additional equipment attachment elements **22** may be attached to the horizontal stabilization member **20** between its first and second ends **20a**, **20b**. For example, the rooftop mount **10** can include third, fourth, fifth, or more equipment attachment elements **22** connected to the horizontal stabilization member **20** between the first and second equipment attachment elements. In some embodiments, unlike the first and second equipment attachment elements **22**, which may be connected to first and second connection members **18**, respectively, these additional equipment attachment elements **22** may be attached to the horizontal stabilization member **20** only and not to any piece of the rooftop mount **10** that is connected directly to the container's roof **16** (i.e., not to any of the connection members **18**). In other embodiments, such additional equipment attachment elements **22** may be attached to additional (i.e., third, fourth, etc.) connection members **18**, wherein the additional connection members are attached to the roof **16** of the container **14**.

In exemplary embodiments, the rooftop mount device **10** is made of steel or other metal. In other embodiments, the rooftop mount **10** may be made from a semimetal like aluminum, an alloy, a plastic or other hydrocarbon-based synthetic material, wood, or any other suitable material that is sufficiently strong, durable, and weather-resistant to store and transport the exercise equipment **12**. In some embodiments, one or more of the various components of the rooftop mount **10** may be constructed from materials that differ from the materials from which the other component or components are constructed.

The invention also relates to a kit that includes the parts of the rooftop mount device **10** described herein above. The components of the rooftop mount device **10** may be provided as parts of a kit that can be assembled to use with a container **14** already in the possession or control of the user. In this way, existing containers may be repurposed using the kit, which contains the rooftop mount device **10** for assembly and mounting onto the roof **16** of such existing container **14**. In the case of kits, only the rooftop mount device and its parts are provided to the user because the user already has his or her own container. As a kit, the rooftop mount device **10** can be disassembled when the user desires to use the container **14** for purposes other than exercise purposes. The stand-alone portable gym kit includes a rooftop mount **10** that can be used to re-purpose any existing container for conversion into a portable gym container system. Such a kit can include all of the necessary fasteners and attachment apparatuses necessary to connect the rooftop mount to a roof of the container. The portable gym kit can also include exercise equipment as described elsewhere herein as well as other fitness equipment such as, for example, weights, balls, ropes, etc., along with storage means to prevent the exercise equipment and any related fitness equipment from moving around inside the container during storage and transport.

The invention also relates to a portable gym system. In an exemplary embodiment, the portable gym system includes the rooftop mount device **10**, exercise equipment **12**, and a container **14**. In other embodiments, the portable gym system may include only the rooftop mount device **10** and a container **14** so that the user must supply his or her own exercise equipment. In still other embodiments, the portable gym system may include only the rooftop mount device **10** and exercise equipment **12** so that the user must supply his or her own container **14**. The portable gym system can include one rooftop mount and corresponding exercise

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equipment for attachment thereto on each side the portable container's roof. For example, when the portable container is shaped as a parallelepiped, the portable gym system can include four rooftop mounts, one for each side of the portable container, although in some embodiments, the system could include fewer than four rooftop mounts (e.g., one, two, or three) for attachment to the roof of the portable container. In other embodiments of the system in which the portable container and its roof include more or less than four sides, the system can include one rooftop mount for each side of the roof or fewer than one per each side of the roof.

As shown in FIGS. 39 and 40, in some embodiments, the system includes horizontal stabilization members 20 installed on four sides of the roof 16 of the shipping container 14 so that all sides of the shipping container include a rooftop mount device 10 attached thereto. In such embodiments, more exercise equipment may be attached so that more users may utilize the system for exercise. The system includes racking that surrounds or encircles the shipping container or other object. The racking includes horizontal connection members 42, equipment connection members 44, and horizontal cross members 46. Each equipment attachment element 22 may have one, two, or more horizontal connection members 42 attached perpendicularly thereto at a first end 42a of each horizontal connection member 42. Each horizontal connection member 42 includes a second end 42b that attaches perpendicularly to an equipment connection member 44. The equipment connection members 44 are oriented vertically and include multiple points of attachment for exercise equipment. Each equipment connection member 44 includes a top end 44a and a bottom end 44b. At the bottom end 44b, each equipment connection member 44 contacts a ground or floor surface. At the top end 44a, each equipment connection member 44 is attached to adjacent equipment connection members to its left and right by horizontal cross members 46. The horizontal cross members 46 provide additional stability to the system and may be used as pull-up bars or for attachment of exercise equipment. In these embodiments, the system includes one or more rooftop mount devices that connect to all sides of the roof of the shipping container or other object so that the shipping container or object and its roof, including its corners, are encircled or surrounded by the one or more rooftop mount devices (and racking) as shown in FIG. 40. Thus, the shipping container or other object is completely and continuously surrounded by the one or more rooftop mount devices (and racking) by attachment to the roof of the shipping container or object.

The portable gym system is transportable from one location to another location. When not in use, for example, during transport, the exercise equipment is disassembled and stored inside the container. When the portable gym reaches a destination where use of the exercise equipment is desired, the exercise equipment is unpacked from the container and configured and assembled as desired. Certain parts of the exercise equipment are mounted to a roof of the container to provide stability and proper height for configuration, assembly, and use of the exercise equipment. The portable gym can be provided as a kit that may be assembled, disassembled, and transported for use, and particularly in locations where access to gym facilities and exercise equipment are otherwise unavailable.

Other Embodiments

It is to be understood that while the invention has been described in conjunction with the detailed description

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thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

What is claimed is:

1. A rooftop mount device for connecting to a roof of a container and attaching exercise equipment thereto, the rooftop mount device comprising:

at least two connection members that are attachable to a roof of a container; wherein the at least two connection members each comprise a vertical portion and a tab protruding from one side, wherein the tab is sized and shaped to fit into an aperture of a corner casting of the container to permit installation of the at least two connection members without tools; and

at least one horizontal stabilization member for connecting between two opposing connection members of the at least two connection members when the rooftop mount device is attached to the roof of the container; wherein one or more items of exercise equipment are connectable to the rooftop mount device at the connection members, at the horizontal stabilization member, or at both; and

wherein the exercise equipment is adjustably attachable both horizontally and vertically at a plurality of connection points on the connection members and the horizontal stabilization member.

2. The rooftop mount device of claim 1, comprising at least four of the connection members.

3. The rooftop mount device of claim 1, wherein each connection member connects to the roof of the container at a corner casting, wherein one corner casting is located at each corner at a top side of the container.

4. The rooftop mount device of claim 1, wherein the horizontal stabilization member comprises a horizontal brace comprising a first end and a second end, wherein the first end attaches to at least one of the connection members.

5. The rooftop mount device of claim 1, further comprising a center brace comprising a left vertical portion, a right vertical portion, and a horizontal portion; wherein the left and right vertical portions each attach to the roof of the container at opposing points located at left and right sides of the container between connection members; and wherein the horizontal portion extends across the roof and comprises a first end that connects to the left vertical portion and a second end that connects to the right vertical portion.

6. The rooftop mount device of claim 5, wherein the left vertical portion and the right vertical portion of the center brace each comprise at least one connection body for attaching the each at least one horizontal stabilization member to the vertical portion of the center brace adjacent to each at least one horizontal stabilization member.

7. The rooftop mount device of claim 5, wherein the center brace comprises a plate for contacting against the roof at one side, wherein the plate extends downward from a bottom of the horizontal portion; and wherein the center brace further comprises a tension bolt that, when tightened, presses the plate against the roof of the container to maintain the at least one horizontal stabilization member in position during use of the exercise equipment.

8. The rooftop mount device of claim 1, wherein the at least two connection members, the at least one horizontal stabilization member, or both comprise a plurality of apertures for receiving pins or other fasteners to connect exercise equipment to them.

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9. The rooftop mount device of claim 1, further comprising one or more equipment attachment elements connected to the at least one horizontal stabilization member, wherein each of the one or more equipment attachment elements is adjustable in its position of attachment to the at least one horizontal stabilization member.

10. A portable gym system comprising:

an object comprising a roof and a storage area for holding and storing one or more items of exercise equipment; a rooftop mount device for connecting to the roof and attaching the exercise equipment thereto, the rooftop mount device comprising:

connection members that are attachable to the roof; wherein the connection members each comprise a vertical portion and a tab protruding from one side, wherein the tab is sized and shaped to fit into an aperture of a corner casting of the container to permit installation of the connection members without tools; and

a horizontal stabilization member for connecting between vertical portions of two opposing connection members when the rooftop mount device is attached to the roof;

wherein the exercise equipment is connectable to the rooftop mount device at the connection members, at the horizontal stabilization member, or at both; and wherein the exercise equipment is adjustably attachable both horizontally and vertically at a plurality of connection points on the connection members and the horizontal stabilization member.

11. The portable gym system of claim 10, further comprising exercise equipment storable in the storage area of the object, wherein at least part of the exercise equipment is attachable to the rooftop mount device.

12. The portable gym system of claim 10, wherein the object comprises a shipping container, a storage container, a trailer of a tractor trailer, a box of a box truck, a trailer home, a motor home, a camper, or a recreational vehicle comprising sufficient height and enclosing an interior space in which exercise equipment is storable.

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13. The portable gym system of claim 10, wherein the system comprises horizontal stabilization members installed on two, three, or four sides of the roof.

14. The portable gym system of claim 10, wherein the system comprises equipment attachment elements attached to at least one horizontal connection member at a first end of each horizontal connection member; wherein the at least one horizontal connection member comprises a second end that attaches perpendicularly to an equipment connection member; wherein the equipment connection member comprises a top end and a bottom end; and wherein at the top end, the equipment connection member is attached to adjacent equipment connection members by horizontal cross members.

15. The portable gym system of claim 10, wherein the system comprises one or more rooftop mount devices that connect to all sides of the roof of the object so that the object and its roof are encircled by the one or more rooftop mount devices.

16. A rooftop mount device for connecting to a roof of a container and attaching exercise equipment thereto, the rooftop mount device comprising:

at least two connection members that are attachable to a roof of a container; wherein each connection member comprises attachment means for connecting one or more items of exercise equipment to the connection member without contacting a wall of the container; wherein the at least two connection members each comprise a vertical portion and a tab protruding from one side, wherein the tab is sized and shaped to fit into an aperture of a corner casting of the container to permit installation of the at least two connection members without tools;

wherein the exercise equipment is connectable to the rooftop mount device at the connection members; and wherein the exercise equipment is adjustably attachable and repositionable both horizontally and vertically at a plurality of connection points on the connection members.

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